

Beyond Prepare-and-Measure: The Cost of Trustless Detectors in Quantum Key Distribution

Linus Kelsey

Aug 2026

Abstract

Quantum Key Distribution (QKD) was first put forward in 1984 by Bennett and Brassard through their foundational proposal for bipartite secure key exchange taking advantage of quantum mechanics, now commonly known as BB84. In the following decades, a number of inspired methods have emerged to guarantee shared secrecy. In particular, Measurement-Device-Independent (MDI) QKD, presented in 2012, improves BB84 security assumptions by eliminating the need to trust photon detectors. In order to be able to deploy QKD at scale, providers must map the trade-offs of either protocol, and comparisons of the two in secure information exchange rate (key rate), cost and scalability allow for efficient application of this maturing technology. In this report, we aim to contribute to the solution of this problem through use of NetSquid, a quantum network discrete event simulation (DES) engine, and an algebraic cost model, to simulate and compare network behaviour for both MDI-QKD and BB84. We see that the price to pay for MDI's improved security is a drop in key rate, of 5 to 25 times depending on the dominant hardware impairment, in modest metropolitan networks under realistic hardware assumptions.

Contents

1	Introduction	4
2	Shared Quantum Secrecy	6
2.1	BB84	6
2.2	Measurement-Device-Independent QKD	7
2.3	Theoretical Performance Bounds	9
2.4	Network Topologies	9
3	Methods	10
3.1	Performance Analysis	10
3.1.1	NetSquid Simulation Engine	10
3.1.2	Hardware Model	11
3.1.3	Bipartite Single-Link Analysis	13
3.1.4	Multi-User Network Analysis	15
3.2	Cost Model	18
3.3	Scalability Analysis	19
4	Results	20
4.1	Point-to-Point Performance	20
4.2	Network Performance	26
4.3	Cost	28
4.4	Deployment Recommendations	32
5	Discussion	34
5.1	The Security, Rate, and Cost Trade-off	34
5.2	Cost Model Limitations	35
5.3	Simulation Model Limitations	36
5.4	Implications for Protocol Selection	38
6	Conclusion	39
6.1	Summary of Contributions	39
6.2	Findings	39
6.3	Future Work	40
A	Additional Point-to-Point Parameter Sweeps	43

B Relay Placement Optimisation	46
C Real-World Deployment Considerations	51
D NetSquid: Limitations and Speedup Strategies	56
E Code and Reproducibility	58

1 Introduction

The advent of large-scale quantum computing poses a fundamental threat to classical public-key cryptography, the class of primitives securing a significant majority of Internet data [1]. Modern key-exchange standards, including RSA [2] and Elliptic Curve Cryptography [3, 4], derive their security from the computational hardness of integer factorisation and the discrete logarithm problem. Shor’s algorithm [5], published in 1997, demonstrated that both problems can be solved in polynomial time on a sufficiently powerful quantum computer, rendering these standards vulnerable to a quantum-capable adversary. Post-Quantum Cryptography (PQC) responds to this threat by identifying classically hard problems not susceptible to known quantum speed-ups; Quantum Key Distribution (QKD) takes the complementary route, using quantum mechanics to guarantee shared secrecy rather than computational assumption.

QKD was first conceived by Wiesner [6] in the 1970s (though his original work was not published until 1983) and given its first operational form by Bennett and Brassard in 1984 [7]. In the decades since, QKD has matured into one of the most field-tested quantum technologies, with national-scale fibre deployments serving over 150 users across thousands of kilometres [8] and metropolitan testbeds demonstrating sustained multi-vendor operation [9, 10]. These demonstrations and others like them have motivated a growing body of research into how QKD protocols should be chosen and deployed at scale. This work focuses on two protocols that represent a natural axis of comparison: BB84, the original Prepare-and-Measure (PM) scheme, and Measurement-Device-Independent QKD (MDI-QKD) [11]. Their protocols and theoretical performance bounds are introduced in Section 2.

MDI-QKD improves upon BB84’s security model by removing the need to trust photon detectors, a difficult assumption to verify in practice that has been exploited in hardware-level attacks [12–14]. This advantage, however, comes at a cost: at metropolitan distances, BB84 consistently delivers rates an order of magnitude higher than MDI under comparable hardware conditions [15, 16]. For network providers selecting between the two protocols, this trade-off must be mapped carefully over variations in distance, topology, and network scale. The question is not simply which protocol is more secure, but under what deployment conditions MDI’s security advantage justifies its rate penalty, and at what economic and growth cost.

Strong commercial momentum behind both protocols has been evidenced by a progression of field deployments from early national testbeds [8–10, 17] through metropolitan and long-haul fibre networks [15, 18] to dedicated MDI-QKD systems [19–21], heterogeneous multi-vendor network trials [22], and European infrastructure programmes [23]. However, the

quantitative basis for choosing between BB84 and MDI-QKD for a given application remains poorly characterised. Simulation offers a low-cost route to this characterisation: hardware parameters can be reconfigured and swept freely, and results interrogated directly, without deploying physical infrastructure. Yet simulation-based studies of metropolitan quantum networks to date, such as Quantum City’s fixed star-topology architecture [24], do not systematically sweep topology, relay placement, and hardware parameters across network scale to support protocol selection. Cost models for MDI networks are largely absent from the literature: whilst integer linear programming approaches have been developed for PM and entanglement-based QKD infrastructure [25], similar work for MDI has not been seen. A rigorous parametric comparison across key rate, cost, and scalability is absent.

This work addresses these gaps through a two-part methodology. First, we implement and simulate both protocols within NetSquid [26], a discrete-event quantum network simulation engine, across a hardware model spanning ideal to physically realistic conditions. Second, we derive deployment cost formulae algebraically using symbolic per-component parameters expressing each protocol’s total infrastructure cost as a closed-form function of user count and topology. Relative capital expenditure (CapEx) comparison reveals the primary analytical result of a structural scaling difference between protocols. NetSquid was chosen over a bespoke analytical or probabilistic model for a reason that outlasts this specific comparison: its component/node/protocol abstraction (Section 3) is general enough to host arbitrary quantum network topologies, hardware models and protocols. Classical networking has long relied on general-purpose discrete-event simulators such as ns-3 [27] and OMNeT++ [28] to plan and evaluate infrastructure at commercial scale; QKD deployment planning has no comparably mature counterpart. A secondary aim of this work is accordingly to lay early groundwork toward one. We compare BB84 and MDI-QKD across three dimensions: key rate as a function of distance and hardware parameters; relative deployment cost as a function of network size; and scalability, defined as the marginal key rate when considering deployment questions for a fixed metropolitan user set. The remainder of this report is structured as follows. Section 2 introduces the protocols, their hardware underpinnings, and the theoretical performance bounds that govern them. Section 3 describes the simulation architecture and cost model. Section 4 presents results across all three comparison axes, discussed in Section 5, and Section 6 concludes with deployment recommendations and directions for future work.

2 Shared Quantum Secrecy

Quantum Key Distribution refers to any communications protocol in which two or more parties use quantum channels, the sharing of quantum states, together with authenticated classical channels and post-processing, to establish a shared secret cryptographic key. The security of such a scheme is guaranteed by the laws of quantum mechanics, providing information-theoretic security against any eavesdropper whose capabilities are bounded by quantum physics. Since Bennett and Brassard’s foundational paper in 1984 [7], numerous variants of QKD have been proposed. This work focuses on two: the original PM scheme (BB84) and MDI-QKD, proposed by Lo, Curty and Qi in 2012 [11].

2.1 BB84

BB84 is representative of the class of PM protocols. Alice begins by preparing a string of n photons, each encoded in one of four polarisation states (0° , 45° , 90° or 135°) selected uniformly at random. In the qubit formalism, these correspond to $|0\rangle$, $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, $|1\rangle$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$, respectively. Alice selects a basis ($\{|0\rangle, |1\rangle\}$ or $\{|+\rangle, |-\rangle\}$) for each qubit uniformly and independently, then uniformly and independently again, she applies a bit flip within that basis before transmission.

She transmits these quantum states to Bob over a quantum channel. Bob measures each photon in a uniformly and independently random basis. Without disturbance to the channel, where Alice and Bob’s bases choices overlap, they have matching bit values. Alice and Bob then use an authenticated classical channel to compare basis choices and discard all photons measured in mismatching bases, producing a raw, sifted key. A portion of this sifted key is compared over the classical channel to estimate the Quantum Bit Error Rate (QBER): the fraction of photon pairs measured in the same basis that nevertheless disagree. A QBER below 11% is taken as evidence that the channel has not been compromised [29]. Alice and Bob then apply error correction and privacy amplification (hash-based post-processing that shortens the raw key, removing any partial information an eavesdropper may have gathered) to produce the final shared secret key. Figure 1 illustrates the basic bipartite logical layer.

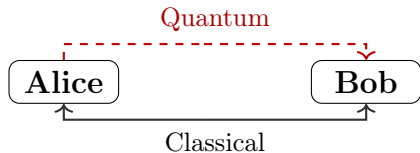


Figure 1: Communication diagram for point-to-point BB84. Alice sends quantum states to Bob; basis reconciliation and post-processing occur over authenticated classical channels.

Security proofs for BB84, and QKD generally, are conditional on the modelling assumptions made about devices, network, and attacker capabilities. Single-photon sources (SPS) have, until recently, been very hard to manufacture, and a particular vulnerability in practice arises from using the cheaper, more easily produced weak coherent pulse (WCP) laser sources, whose photon number per pulse follows a Poisson distribution. Multi-photon pulses allow a Photon Number Splitting (PNS) attack [30], in which Eve intercepts and holds trailing photons until bases are announced, extracting key information without disturbing the QBER. The decoy-state technique [31] mitigates this by varying pulse intensity randomly, enabling Alice and Bob to tightly bound the single-photon contribution to their key. In the present work, we model photon sources as SPSs.

2.2 Measurement-Device-Independent QKD

MDI-QKD addresses a fundamental vulnerability shared by all PM schemes: the need to trust the measurement devices at Bob’s node. Side-channel attacks on detectors represent a significant practical threat, as the gap between idealised device models assumed in security proofs and real hardware is difficult to close. MDI removes detectors entirely from the trust boundary by introducing an untrusted third-party relay, Charlie, who performs all measurement operations.

In MDI-QKD, both Alice and Bob independently prepare photon strings in the manner of BB84 and transmit these to Charlie. Charlie performs a Bell State Measurement (BSM) on each arriving photon pair, projecting onto the two-qubit Bell basis:

$$|\phi^\pm\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}, \quad |\psi^\pm\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}.$$

The physical mechanism underlying the BSM is Hong-Ou-Mandel (HOM) interference [32]: two indistinguishable photons arriving simultaneously at a 50:50 beam splitter always exit the same output port due to bosonic quantum interference. Coincident clicks at the downstream polarising beam splitters then identify the projected Bell state. In this BSM implementation, only two of the four Bell states ($|\psi^+\rangle$ and $|\psi^-\rangle$) are unambiguously distinguishable [11]; detection events corresponding to $|\phi^\pm\rangle$ are discarded as inconclusive. Charlie announces the measurement outcome over a classical channel to both Alice and Bob. Since the BSM measures only the *joint correlation* between the two incoming photons rather than their individual polarisations, an eavesdropper with full control of Charlie’s apparatus can learn nothing about Alice’s or Bob’s individual bit values. Photon indistinguishability is therefore a prerequisite for MDI-QKD: timing jitter, spectral mismatch, and polarisation misalignment all reduce HOM visibility and degrade BSM success probability. The bipartite logical layer

for MDI-QKD is shown in Figure 2.

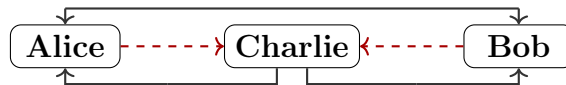


Figure 2: Communication diagram for point-to-point MDI-QKD. Both Alice and Bob transmit quantum states to Charlie; Charlie performs a BSM and announces the result classically to both parties.

Alice and Bob sift by matching bases as in BB84. Upon receiving the BSM outcome, each party knows their own basis choice and bit value, and can deduce the other’s bit via the outcome. Depending on the Bell state announced, one (pre-agreed) party selectively bit-flips to ensure key agreement, as summarised in Table 1. Privacy amplification and error correction then proceed as in BB84.

BSM outcome	$ \psi^+\rangle$	$ \psi^-\rangle$
Diagonal basis	No bit flip	Bit flip
Rectilinear basis	Bit flip	Bit flip

Table 1: Bit-flip rules upon receipt of BSM outcome [11]. Alice and Bob agree in advance which party applies the flip; if both flip, there is no net change.

The abstract operation of the BSM in quantum information terms is shown in Figure 3: a CNOT gate followed by a Hadamard and measurement. The physical implementation uses a 50:50 beam-splitter and two polarising beam-splitters on each arm, with the Hong-Ou-Mandel effect ensuring that only coincidence detection events, two simultaneous clicks at the detector array, constitute valid BSM outcomes.

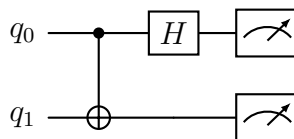


Figure 3: Bell State Measurement circuit: CNOT gate followed by Hadamard, then measurement in the computational basis.

MDI-QKD’s security advantage comes at a cost. Key generation requires a successful coincidence detection at Charlie – both photons from Alice and Bob must arrive and be detected within the same time window. This requirement, combined with independent connection losses on each arm, imposes a rate penalty relative to BB84’s single-link loss [11]. MDI’s QBER tolerance is also more sensitive than BB84’s, as two independent quantum channels each contribute noise, enlarging the attack surface accounted for by security proofs.

2.3 Theoretical Performance Bounds

The maximum achievable secure key rate in any point-to-point (no quantum repeater) QKD scheme is bounded by the Pirandola–Laurenza–Ottaviani–Banchi (PLOB) bound [33]. This fundamental result establishes that, against all attacks permitted by quantum physics, the key rate of a repeater-less link scales at most linearly with channel transmittance η :

$$R \leq -\log_2(1 - \eta) \approx \eta \log_2 e \quad \text{for } \eta \ll 1. \quad (1)$$

Since transmittance along all channels decays exponentially with length L as $\eta = 10^{-\alpha L/10}$ (with α the fibre attenuation coefficient in dB/km), the PLOB bound implies that key rates must fall exponentially with distance. Both BB84 and standard MDI-QKD are subject to this bound.

Twin-Field QKD (TF-QKD), proposed in 2018 by Lucamarini et al. [34], exploits the same untrusted-relay tenet as MDI but replaces the two-photon BSM with single-photon interference. The result is that key rates scale as $\mathcal{O}(\sqrt{\eta})$ rather than $\mathcal{O}(\eta)$, exceeding the PLOB bound and extending the distance at which secure communication is achievable. TF-QKD thus represents the most practically significant near-term stepping stone beyond the repeater-less limit, though it falls outside the scope of this work.

2.4 Network Topologies

When QKD is deployed beyond a single point-to-point link, network topology becomes a primary driver of cost and scalability. BB84, in the form considered here, requires a dedicated quantum channel between every pair of users: for N users, a full mesh demands $\mathcal{O}(N^2)$ fibre spans, each carrying an independent quantum signal. MDI-QKD’s architecture naturally admits a star topology instead: all users direct their pulses to a shared, untrusted measurement node (Charlie), which performs the Bell-state measurement without accessing the key material. For K relay nodes each serving a cluster of users, the MDI network as introduced in Section 3 requires $\mathcal{O}(N)$ user–relay fibre spans and $\mathcal{O}(K^2)$ inter-relay backbone links, both substantially below the $\mathcal{O}(N^2)$ full mesh for $K \ll N$. This structural asymmetry is the central economic and scalability argument evaluated in this work.

3 Methods

We compare BB84 and MDI-QKD across three dimensions: performance, deployment cost, and scalability, for both point-to-point (P2P) and network settings. Together these axes capture the trade-off a network provider faces when selecting between the two protocols for a given metropolitan deployment.

Performance. The primary metric is sifted key rate, measured in bits per second: the length of the matched-basis sifted key divided by the time taken to produce it. Since all QKD-generated keys are information-theoretically equivalent in security regardless of protocol, key rate is a core performance discriminant between BB84 and MDI-QKD. We simulate key rate as a function of distance and deployment configuration using NetSquid [26], a discrete-event quantum communication simulation engine, across a ten-layer hardware model that progressively introduces realistic physical imperfections.

Cost. Deployment cost is modelled algebraically using symbolic per-component parameters. The count of each component is determined by the network topology under each protocol. Since quantum optic manufacturer prices are generally not publicly available, we focus on the *relative* cost structure between protocols.

Scalability. We define scalability as the marginal key rate lost when an existing network has its topology modified. Since the two protocols route traffic differently by nature, they exhibit fundamentally different scaling behaviour. We quantify this by sweeping deployment variables at fixed topology parameters and extracting per-user key rates.

3.1 Performance Analysis

The primary performance metric is the sifted key rate R , measured in bits per second: the key length extracted by a simulation run divided by the simulation duration. To reflect QKD conditions for information-theoretic security, a symmetric QBER threshold of 11% is applied: simulation runs with QBER above this value are discarded as failures [29]. All key rates are computed in the asymptotic regime, without finite-key corrections, error correction, or privacy amplification. R is therefore a sifted key rate, not a secure key rate (discussed in more detail in Section 5), simplifying the modelling.

3.1.1 NetSquid Simulation Engine

NetSquid [26] is a discrete-event simulation (DES) engine for quantum networks developed at QuTech, chosen for this work for three reasons. First, NetSquid natively tracks qubit states as density matrices and exposes standard gate operations and projective measurement,

enabling BB84 encoding and BSM’s to be implemented directly as quantum operations. Second, the DES clock provides per-trial simulation timestamps to nanosecond precision, making key rate computable from the elapsed simulation time between photon emission and protocol completion; propagation delay is introduced naturally by channel parameters. Third, the `NodeProtocol` abstraction models each party’s logic as an event-driven coroutine that *waits* for port inputs; this allows Alice, Bob, and Charlie’s asynchronous exchanges to be written sequentially without explicit thread management, and maps directly onto the protocol descriptions in Section 2.

Further, NetSquid organises a simulation into three layers of abstraction. *Components* represent physical hardware: `QuantumChannel` objects carry photons between nodes, applying noise models as Kraus operators; `ClassicalChannel` objects carry classical messages for basis reconciliation and BSM announcements. *Nodes* represent network parties (Alice, Bob, Charlie), each holding named ports through which components connect; nodes own their local quantum processors and memory registers. *Protocols* are event-driven coroutines that run on nodes, yielding control to the DES scheduler until a subscribed event fires. This three-layer structure is modular: changing a component’s noise model is localised to a single object and does not require protocol logic to change. The same separation is what makes NetSquid a plausible foundation for a general-purpose QKD network simulator: adding a further protocol, hardware model, or topology does not require restructuring the engine underneath, and this is the same design principle underlying commercial-grade classical network simulators such as ns-3 [27] and OMNeT++ [28]. NetSquid’s computational cost, and the extent to which this general-purpose engine is well suited to network-scale QKD simulation, is discussed in Appendix D.

3.1.2 Hardware Model

Physical impairments are introduced via a ten-layer parametric model. Each layer adds one impairment cumulatively to the previous, from a fully ideal baseline (layer 0) to a system incorporating all realistic imperfections (layer 9). This structure allows the marginal impact of each physical parameter on key rate to be isolated and forms the basis of the parameter sweep analysis in Section 4. Table 2 lists the ten layers.

Layer 0 is the ideal baseline: photons propagate without loss or noise, so key rate is limited only by propagation delay. Bipartite sweeps at this layer vary distance across the metropolitan range (0–50 km), establishing an upper bound on achievable key rate and confirming correct simulation timing before physical impairments are introduced.

Layer 1 adds fibre attenuation, α : in the simulator, each photon is discarded before transmission with probability $p_{\text{loss}} = 1 - 10^{-\alpha L/10}$, in line with the Beer-Lambert model [42],

Layer	Parameter added	Symbol	Default value	Verification source
0	Ideal baseline	—	All perfect	—
1	Fibre attenuation	α	0.18 dB/km	Corning [35]
2	Detector efficiency	η_Z	0.90	ID Quantique [36]
3	Dark count rate	d_c	50 cps	ID Quantique [36]
4	TX insertion loss	L_i	0.10 (10%)	OZ Optics [37]
5	RX node loss	L_n	2.0 dB	Duplinskiy <i>et al.</i> [38]
6	Source error rate	ε_s	0.015	Quandela [39]
7	Fibre dephasing	β	$3.2 \times 10^{-7} \text{ km}^{-1}$	Corning [35]
8	X-basis detector bias	η_X	0.715	Grasselli <i>et al.</i> [40]
9	Beam-splitter efficiency	η_{BS}	0.97	Thorlabs [41]
—	Relay position	δ	0.5	User symmetry

Table 2: Ten-layer hardware model. Layer k includes all impairments of layer $k - 1$ plus the new parameter listed. Layer 9 is the realistic operating point used for network simulations.

where L is the arm length.

Layer 2 introduces Z-basis detector efficiency $\eta_Z = 0.90$: each photon arriving at the receiver is accepted with probability η_Z .

Layer 3 adds dark count rate $d_c = 50$ cps: a spurious click fires per time slot with probability $d_c/(d_c + f)$, where f is the source repetition rate, contributing to QBER.

The default detector parameters ($\eta_Z = 0.90$, $d_c = 50$ cps, layers 2–3) correspond to a superconducting nanowire single-photon detector (SNSPD) operating point, verified against the ID Quantique (IDQ) ID281 [36]. The bipartite parameter sweeps in Section 4.1 additionally span the single-photon avalanche diode (SPAD) regime ($\eta_Z \approx 0.10$ – 0.25 , $d_c \approx 100$ – 200 cps), representative of the IDQ ID230 [43], covering the full range of detector technologies relevant to near-term metropolitan deployments.

Layer 4 adds transmission (TX) insertion loss $L_i = 0.10$: a fixed fraction of photons is lost at the source before entering the fibre, absorbed into the combined transmittance $T = 10^{-\alpha L/10}(1 - L_i)$. This models a polarisation-maintaining fused fibre coupler combining two orthogonally-polarised branches (e.g. H/V or $\pm 45^\circ$) into a single output fibre for state preparation.

Layer 5 introduces receiver (RX) node loss $L_n = 2.0$ dB as a single lumped parameter applied uniformly to both protocols, capturing the insertion loss of optical receiver components. Each photon surviving fibre transmission is discarded with probability $1 - 10^{-L_n/10}$ [42] before reaching the detector. Individual component losses are not modelled separately and the protocol-specific breakdown is noted as a limitation in Section 5.

Layer 6 adds source error rate $\varepsilon_s = 0.015$: each emitted photon receives an unintended $\hat{X}$ flip with probability ε_s , modelling multi-photon emission events characterised by $g^{(2)}(0)/2$ [44]

for the reference single-photon source [39]. Taking the standard approximation that half of such multi-photon events go undetected as errors, $\varepsilon_s \approx g^{(2)}(0)/2$. Ideal single-photon sources are assumed throughout; WCP sources and decoy-state techniques [31] are not modelled, so reported rates represent an upper bound relative to practical laser deployments.

Layer 7 adds fibre dephasing $\beta = 3.2 \times 10^{-7} \text{ km}^{-1}$: NetSquid’s `DephaseNoiseModel` applies a per-photon dephasing error with probability $p_\phi = \min(1, \beta L)$ proportional to arm length, modelling polarisation-mode dispersion (PMD) in standard single-mode fibre. Mean differential group delay $\Delta\tau = D_{\text{PMD}}\sqrt{L}$ [42] induces a polarisation-dephasing probability $p_\phi \approx (\Delta\tau/T)^2/2$, with $T = 50 \text{ ps}$ the detector timing-discrimination window (IDQ ID281 jitter [36]). Since $\Delta\tau^2 \propto L$, p_ϕ is linear in L , so dephasing is a constant per-km rate $\beta = D_{\text{PMD}}^2/(2T^2)$.

Layer 8 introduces X-basis detector bias η_X : photons measured in the X basis are detected with lower efficiency than those in the Z basis, reflecting the extra optical stage (dispersive medium/interferometer) required for the test-basis measurement. This value follows Grasselli *et al.* [40], who model the test-basis (X) measurement’s extra optical stage as an additional 1 dB insertion loss, giving $\eta_X = \eta_Z \times 10^{-1/10}$. Since insertion loss is a passive multiplicative factor, this ratio is independent of the detector’s own quantum efficiency.

Layer 9 (MDI only) adds beam-splitter efficiency η_{BS} : the HOM coupler at Charlie’s relay has non-unit splitting ratio, reducing coincidence detection probability per arm. This value is derived from the Thorlabs TN1550R5F2 fibre-coupled 50:50 coupler [41], a 1550 nm narrowband fused coupler specified with excess loss $< 0.15 \text{ dB}$: $\eta_{\text{BS}} = 10^{-0.15/10} \approx 0.97$ [42]. The relay position $\delta = 0.5$ (Charlie at the link midpoint) is held fixed for all bipartite sweeps, reflecting user symmetry; it is instead computed per-pair from physical fibre lengths in the network setting (Section 3.1.4).

3.1.3 Bipartite Single-Link Analysis

The bipartite simulation sweeps one hardware parameter at a time across a single pair of communicating parties, holding all others at their layer-9 defaults (Table 2). For each parameter value, $r = 1000$ independent Monte Carlo trials are run, bounding the worst-case relative standard error of the mean (SEM), $\text{CV}/\sqrt{r}$ (where CV, the coefficient of variation, is the standard deviation divided by the mean), to $\sim 1\%$ at the 20 km layer-9 operating point (CV $\approx 30\%$, MDI) – comfortably below the smallest quantified sweep effect of 5.1%. Each trial constitutes one complete QKD exchange from photon emission through key sifting. The mean key rate is reported as a function of the swept parameter.

BB84 simulation. Alice and Bob are represented as `Node` objects connected by one quantum channel (Alice $\rightarrow$ Bob) and two classical channels for basis announcement and

sifting. Alice prepares $n = 1,024$ photons at $f = 10^7$ Hz. Each photon is assigned an independent random basis (Z or X) and bit value. Encoding is applied qubit-by-qubit: a bit-1 value applies $\hat{X}$; an X -basis choice applies a Hadamard gate. Source imperfection (ε_s) is modelled by an additional $\hat{X}$ flip applied with probability ε_s , consistent with the multi-photon emission rate $g^{(2)}(0)/2$ of the reference single-photon source. Photons are discarded before transmission with probability

$$p_{\text{loss}} = 1 - 10^{-\alpha L/10} (1 - L_i), \quad (2)$$

modelling both fibre attenuation and TX coupling loss. The quantum channel applies an additional per-photon dephasing error of probability βL . At Bob, each arriving photon is further subject to receiver node loss (probability $1 - 10^{-L_n/10}$), then detected with probability η_Z (Z basis) or η_X (X basis). Dark counts fire with per-slot probability $d_c/(d_c + f)$; a dark click on a slot where a real detection also occurs overrides with a random outcome. Dark clicks can also fire on slots where the photon was lost.

The sifted key is extracted as the list of Bob's raw measurement outcomes at matching-basis slots. QBER is the fraction of bit disagreements between Alice's and Bob's sifted keys:

$$\text{QBER} = \frac{1}{\ell} \sum_{k=1}^{\ell} \mathbf{1}[a_k \neq b_k], \quad (3)$$

where ℓ is the sifted key length. The QBER is measured over the full sifted key, not a sacrificed subset, since no eavesdropper is modelled. Key rate is then:

$$R = \frac{\ell}{T_{\text{end}} - T_{\text{start}}} \quad (\text{bps}), \quad (4)$$

where T_{end} is the simulation time at which Bob completes basis reconciliation and T_{start} is the photon emission start, both in nanoseconds. Monte Carlo runs with QBER > 11% are excluded and counted as failures. No error correction or privacy amplification is applied; R is therefore a sifted key rate rather than a secure key rate, and overestimates the latter (see Section 5).

MDI simulation. Three nodes are instantiated: Alice, Bob, and Charlie. Two quantum channels (Alice $\rightarrow$ Charlie, Bob $\rightarrow$ Charlie) and six classical channels (BSM announcement and basis disclosure) are constructed. Each end node prepares $n = 512$ photons: summed across MDI's two sources, this matches BB84's single-source photon-emission budget, though the resulting per-source pulse-train window is half as long (Section 5.3). Encoding, source error, and per-photon transmission loss are applied identically to the BB84 case, with arm

lengths $L_A = L\delta$ and $L_B = L(1 - \delta)$ where δ locates Charlie on the link; each arm applies an independent dephasing model proportional to its length.

At Charlie, photons arriving from Alice and Bob on the same pulse index are paired. Both photons undergo node coupling loss (L_n) and are each detected with probability $\eta_{BS} \eta_Z$ or $\eta_{BS} \eta_X$ (HOM beam-splitter efficiency $\times$ detector efficiency). A CNOT gate followed by a Hadamard on the control qubit and measurement (a, b) of both qubits implements the Bell state measurement. The four computational outcomes correspond to the four Bell states:

$$(a, b) \mapsto \begin{cases} |\Phi^+\rangle & a = 0, b = 0 \\ |\Phi^-\rangle & a = 1, b = 0 \\ |\Psi^+\rangle & a = 0, b = 1 \\ |\Psi^-\rangle & a = 1, b = 1 \end{cases} \quad (5)$$

Consistent with the linear-optical BSM constraint [11], only $|\Psi^\pm\rangle$ outcomes ($b = 1$) are treated as conclusive; $|\Phi^\pm\rangle$ outcomes ($b = 0$) are discarded. This yields a maximum BSM success probability of 50% for coincident pairs. Dark count clicks on unpaired or lost slots generate spurious conclusive outcomes, contributing to QBER. Charlie announces BSM outcomes and basis-mismatch indices to Alice and Bob; end users discard inconclusive and basis-mismatched rounds. Bob applies basis-dependent bit corrections according to the BSM outcome to restore key agreement with Alice, as in Table 1. The sifted key, QBER, and key rate are computed as per Equations 3–4, with $T_{\text{end}} = \max(T_{\text{end}}^A, T_{\text{end}}^B)$.

3.1.4 Multi-User Network Analysis

The multi-user simulation models a metropolitan network of N users within an $L \times L$ km service area. Users are placed uniformly at random within the grid.

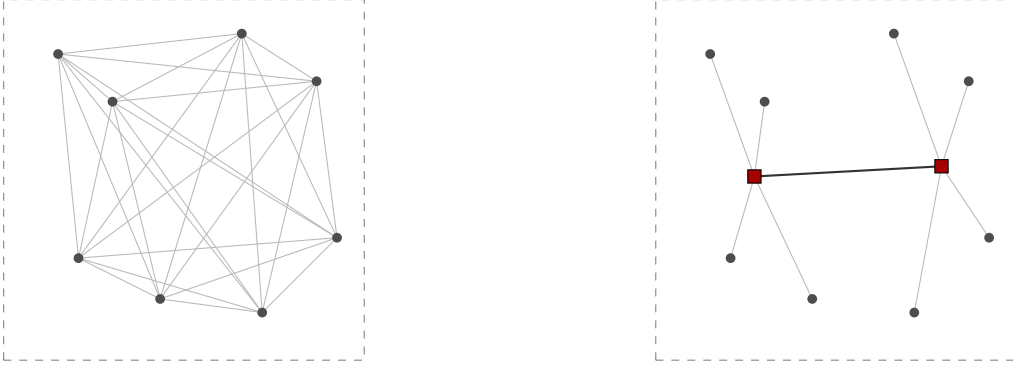
This work focuses on metropolitan-scale networks, where inter-node distances are short enough for both BB84 and MDI-QKD to establish keys at viable rates. Two network protocols are simulated:

BB84 mesh. Every user pair (i, j) communicates over a dedicated direct dark-fibre link of physical length d_{ij} . All pairs are dispatched to the multiprocessing pool simultaneously.

MDI network. For a given set of N users and K relays, each user is connected to and served by its nearest relay. Relays are further connected to one another over a fully-meshed backbone (Figure 4b). A BSM between nodes in the same cluster takes place at their shared relay; a cross-cluster pair instead routes via the backbone – one party’s photon travels to its own relay, and is then optically switched along the backbone to the other party’s relay, where the BSM occurs – in a hub-and-spoke, optical-switch-routed architecture inspired by

commercial MDI-QKD systems such as Q*Bird’s Falcon series [45].

A third protocol, trusted-node BB84 (TBB84), is implemented but not exercised in this work’s core key-rate simulation results, and is included here for completeness. TBB84 uses the same relay architecture as MDI, and routes a key between non-cluster-connected users via one or more relay hops [46]. Standard BB84 occurs on each leg of the path between nodes and classically XOR-combining the resulting per-hop keys along the path allows users to share a secret key. Every relay passed through handles plaintext-equivalent key material, so its security rests on relay site hardware trust unlike MDI’s device-independence (Section 5.1). Appendix C quantifies its deployment cost (identical to MDI’s under this work’s cost model) on a real topology, and discusses its qualitative trade-offs against MDI in reference to a real QKD deployment [47].



(a) BB84 direct mesh: $\binom{8}{2} = 28$ dedicated links. (b) MDI multi-relay: 8 spokes + backbone link.

Figure 4: Example network topology for the same $N = 8$ user configuration under BB84 (left) versus MDI-QKD with $K = 2$ relays (right, relay stations shown as squares). BB84 requires a dedicated fibre for every user pair ($\mathcal{O}(N^2)$ links); MDI instead routes each user to its nearest relay, with the two relays joined by a shared backbone link ($\mathcal{O}(N)$ spokes plus $\mathcal{O}(K^2)$ backbone links).

Relay placement. Placing the K MDI relay nodes is a facility-location problem: the placement minimising deployment cost is the one minimising total fibre length – the geometric median of each cluster, not the k -means centroid (centre of mass of k -means clusters), which instead minimises squared distance. For $K > 1$ fully-meshed relays (Table 3), the backbone term couples all relay positions together, so total fibre length is

$$L(x_1, \dots, x_K) = \underbrace{\sum_{k=1}^K \sum_{i \in C_k} \|x_k - p_i\|}_{\text{spoke cost}} + \underbrace{\sum_{k < l} \|x_k - x_l\|}_{\text{backbone (mesh) cost}}, \quad (6)$$

and per-cluster geometric medians, which ignore this backbone term, are generally suboptimal. The joint optimum instead follows the generalised (multi-facility) Weiszfeld fixed-point

update [48],

$$x_k^{(t+1)} = \frac{\sum_{i \in C_k} \frac{p_i}{\|x_k^{(t)} - p_i\|} + \sum_{l \neq k} \frac{x_l^{(t)}}{\|x_k^{(t)} - x_l^{(t)}\|}}{\sum_{i \in C_k} \frac{1}{\|x_k^{(t)} - p_i\|} + \sum_{l \neq k} \frac{1}{\|x_k^{(t)} - x_l^{(t)}\|}}, \quad (7)$$

updating all K relays simultaneously, with convergence to the joint optimum guaranteed by convexity; for $K = 1$ this reduces to the single-facility case of [48], since the backbone term vanishes. Cluster assignment is initialised by k -means and re-evaluated by nearest-relay distance after each position update: membership is reassigned to convergence for the relay count sweep (Section 3.3), but held fixed for the user count sweep and the placement comparison below. Appendix B derives Equation 7 from first principles and quantifies its advantage over k -means across synthetic topologies, illustrated directly in Figure 16.

Installed cable is longer than straight-line distance in practice: each physical cable of a network’s full graph (inclusive of relay backbone) independently draws a tortuosity factor $\tau \sim \max(1, \mathcal{N}(\mu_\tau, \sigma_\tau^2))$. $\mu_\tau = 1.2$ and $\sigma_\tau = 0.1$ are assumed to represent urban duct routing, scaling both key rate, via increased link loss, and installed-fibre cost (Section 3.2).

To decouple topology variability from Monte Carlo noise, all network experiments are repeated across S independently drawn user placements (seeds), with relay positions re-optimised per seed. Aggregate statistics across seeds produce error bars that characterise topology uncertainty rather than the sampling noise of a fixed topology. Seed count follows a fixed budget: at least 1000 Monte Carlo samples per data point overall, at least 200 per seed – adequate since between-seed CV ($\sim 21\%$, Table 4) dominates the per-pair SEM at $r = 200$ ($\sim 2\%$) by an order of magnitude, so error bars stay seed-limited rather than trial-limited.

Each pair’s Charlie position δ is computed directly from the physical fibre length of each arm – accounting for relay assignment, tortuosity, and, for cross-cluster pairs, the additional backbone hop – rather than fixed at the bipartite sweep’s midpoint value. An optical switch at each relay routes backbone photons to the BSM station for cross-cluster pairs. Its insertion loss is set to 0 dB in the current model; the asymmetry between switch installation cost (per relay) and switch-induced loss (cross-cluster pairs only) is noted as a simulation limitation in Section 5.

For both protocols, every one of the $N(N - 1)/2$ user pairs in a given topology is simulated independently with r Monte Carlo trials; the network key rate reported throughout this work is the mean pairwise point-to-point key rate, averaged first over trials within a pair, then over all pairs, then over S independently drawn topology seeds. This is not a live traffic simulation: no channel contention, scheduling, or multiplexing is modelled. Every user pair is assumed to extract key simultaneously and independently at full channel capacity, with no

interference from other pairs sharing physical infrastructure. The reported metric therefore constitutes a semi-probabilistic geometric upper bound on network throughput: *geometric* because the rate for each pair is determined by the physical path geometry (distances, relay placement, tortuosity); *probabilistic* because relay positions and cable tortuosities are drawn stochastically and averaged across seeds; and an *upper bound* because real deployments would see contention for shared fibre, detector dead time across concurrent channels, and scheduling overhead that this model does not capture. The multi-seeding procedure described prior quantifies the geometric component of this bound across the distribution of plausible topologies for a given N , K , and service area.

3.2 Cost Model

We model the total deployment cost of a given QKD network algebraically using three symbolic per-component parameters: c_s (cost per photon source), c_d (cost per single-photon detector), and c_f (cost per km of installed dark fibre), since manufacturer prices for quantum dot photon sources and single-photon detectors are in general not publicly available. All costs are CapEx only; operational expenditure (OpEx) – in particular cryogenic cooling required by SNSPDs and QD sources – is not modelled. Classical communication infrastructure is assumed to be a sunk cost of existing network infrastructure. The most directly related prior cost analysis is that of Karavias *et al.* [25], which applies integer linear programming (ILP) to PM-QKD and entanglement-based infrastructure planning with concrete component costs; MDI-QKD is not considered there. We instead take a closed-form algebraic approach to the cost structure of BB84 and MDI; the resulting cost formulae, crossover analysis, and commercial packaging discussion are presented alongside the results they inform in Section 4.3.

The two network topologies impose fundamentally different hardware structures. In BB84, every user pair requires a dedicated direct fibre link: the network is a complete mesh of $N(N-1)/2$ quantum channels. At each receiver, a polarising beam splitter (PBS) separates Z and X basis outputs to two SPDs, and an electro-optic modulator (EOM) performs active basis selection. No relay infrastructure is required.

In MDI-QKD, users transmit photons to BSM relay stations rather than to each other. Each relay is a passive station comprising a 50:50 HOM beam splitter, two PBS (one per arm), four SPDs, and one optical switch for routing photons from cross-cluster pairs. To simplify the cost modelling, we package these as a single detector ‘unit’. The network requires N user-to-relay fibre links and $K(K-1)/2$ relay-to-relay backbone links. Table 3 summarises the component counts for both protocols.

Every fibre link’s installed cost is linear in its length via c_f , and fibre length itself scales as $\mathcal{O}(N^2)$ for BB84’s complete mesh versus $\mathcal{O}(N)$ for MDI at fixed relay count K (Table 3)

Component	BB84	MDI
Single-photon sources	N	N
Single-photon detectors	$2N$	$4K$
50:50 beam splitter (HOM)	0	K
Polarising beam splitter	N	$2K$
Electro-optic modulator	N	0
Optical switch	0	K
Fibre links	$N(N-1)/2$	$N + K(K-1)/2$

Table 3: Hardware component counts for a network of N users and K relay nodes. MDI relay nodes are passive BSM stations. BB84 forms a direct mesh with no relay infrastructure.

– the structural cost incentive for relay-based architectures as user counts grow, and the difference the full cost formulae of Section 4.3 build on to derive a cost crossover user count N^* . Installed fibre length additionally reflects the per-link tortuosity model of Section 3.1.4.

3.3 Scalability Analysis

Scalability is characterised through two orthogonal sweeps.

User count sweep. An initial graph of $N_{\min}$ nodes is generated randomly, and users are added one at a time from a single random stream: each new user joins a uniformly-random existing relay catchment and is placed by an isotropic Gaussian around that catchment’s centre. Relay positions are placed once, for the $N_{\min}$ -user network, by the Weiszfeld iteration of Equation 7, and then held fixed as N grows to $N_{\max}$. Catchment membership, once nodes are assigned in the first instance, is thus never reassessed. This models a provider that sizes relay infrastructure for an initial deployment and serves subsequent organic growth without continuously re-optimising it, rather than a provider that instantly, and expensively, re-plans infrastructure after every new customer. For each N , both protocols are simulated and their mean key rate is recorded. The algebraic cost model (Section 3.2) is evaluated over the same sweep, to compare how MDI’s sunk relay hardware and BB84’s per-user direct-fibre requirement each shape marginal CapEx as N grows.

Relay count sweep. User count N is fixed and K is varied from 1 to $K_{\max}$, re-placing relays at each sweep point – modelling a provider scoping an entirely new deployment for a fixed user set. BB84 is relay-independent and appears as a constant baseline. This sweep tests whether the shorter mean user-to-relay distance at higher K outweighs the added relay-to-relay backbone cost, and identifies the performance-optimal relay count for a given N and area.

4 Results

4.1 Point-to-Point Performance

Eleven independent parameter sweeps characterise bipartite BB84 and MDI-QKD key rate around a common realistic metropolitan operating point set by the values in Table 2, with a single parameter varied at a time. Each data point is the mean sifted key rate over 1000 Monte Carlo runs; shaded bands show ± 1 standard deviation of the log-transformed key rate, exponentiated back onto the linear scale shown (hence asymmetric about the mean). Four sweeps that isolate the dominant geometric and hardware effects are discussed here; the remaining seven are reproduced in Appendix A. Where a sweep varies the channel transmittance, the PLOB bound of Section 2.3 is presented.

Distance. Figure 5 sweeps Alice–Bob separation from 1–100 km at fixed attenuation.

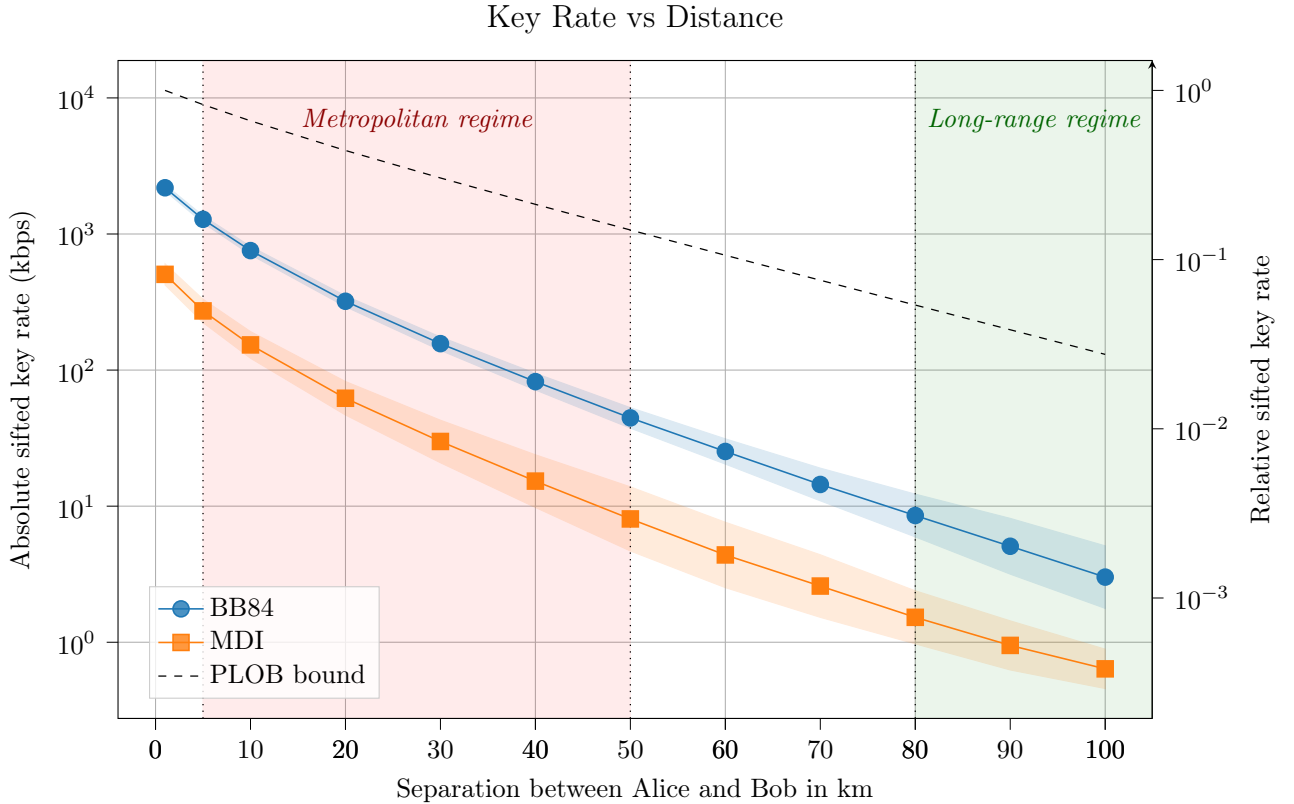


Figure 5: Absolute and relative sifted key rate versus Alice–Bob separation. The PLOB bound is overlaid on the left axis. Shaded vertical bands mark the metropolitan (5–50 km) and long-range (> 80 km) regimes.

The BB84/MDI ratio stays close to a factor of five throughout: $4.3\text{--}5.5\times$ over the full 1–100 km range, even as key rate itself falls by nearly three orders of magnitude. Both protocols

fall increasingly below the PLOB bound with distance: $5.2\times/22.5\times$ at 1 km, widening to $43\times/204\times$ by 100 km for BB84 and MDI, respectively. This provides the practical motivation for the range-extending architectures discussed in Section 5.1 and revisited in Section 6. This is an offset, not a divergence in decay rate: BB84’s key rate scales linearly in η (Section 5.1), the same high-loss regime the PLOB bound itself approaches ($R_{\text{PLOB}} \approx \eta \log_2 e$, Equation 1), so on a logarithmic axis the BB84 curve runs near-parallel to PLOB at long distance. MDI’s roll-off matches BB84’s rate almost exactly (2.86 vs. 2.90 decades over 1–100 km), since both protocols use the same length of fibre and so link loss scales identically.

Detector- and receiver-side parameters. Figure 6 sweeps detector efficiency η_d from 1.0 down to 0.10, shaded to mark the efficiency bands typical of SNSPD (0.80–0.95, IDQ ID281) and SPAD (0.10–0.25, IDQ ID230) technology, and Figure 7 sweeps receiver node/connector loss from 0–6 dB.

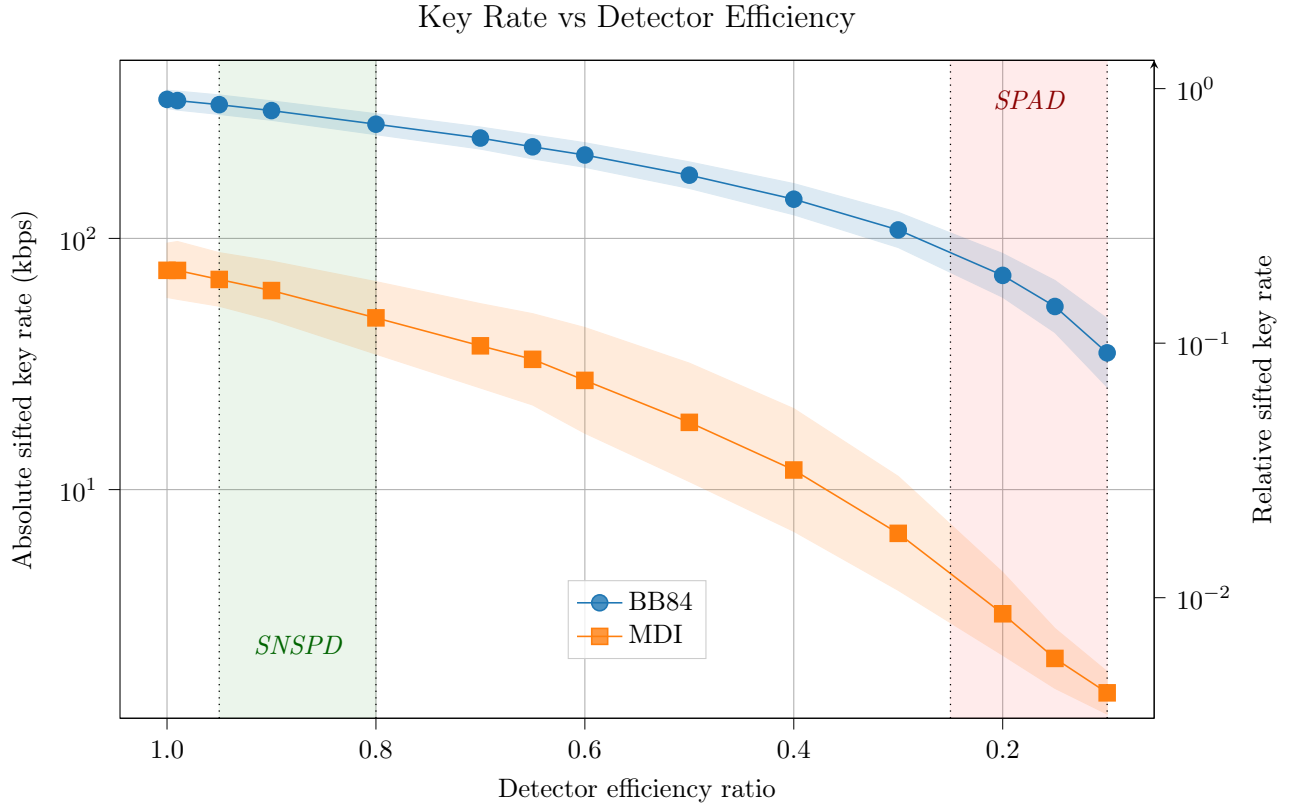


Figure 6: Absolute and relative sifted key rate versus detector efficiency η_d , at fixed 20 km separation and 0.18 dB/km attenuation. Shaded bands mark typical SNSPD (IDQ ID281) and InGaAs SPAD (IDQ ID230) operating ranges.

Unlike distance, the BB84/MDI ratio is not stable under these sweeps: it widens from $4.8\times$ to $25.2\times$ across the bulk of the detector-efficiency range ($\eta_d = 1.0 \rightarrow 0.15$); and from

$3.3\times$ to $13.1\times$ under receiver loss (0–6 dB). Both parameters act *per arm*: detector efficiency and receiver loss enter MDI’s coincidence probability twice, quadratically, but BB84’s detection probability only once (Section 5.1), so any degradation in either hurts MDI disproportionately: detector choice (SNSPD vs. SPAD) therefore matters far more for MDI deployments than for BB84 ones.

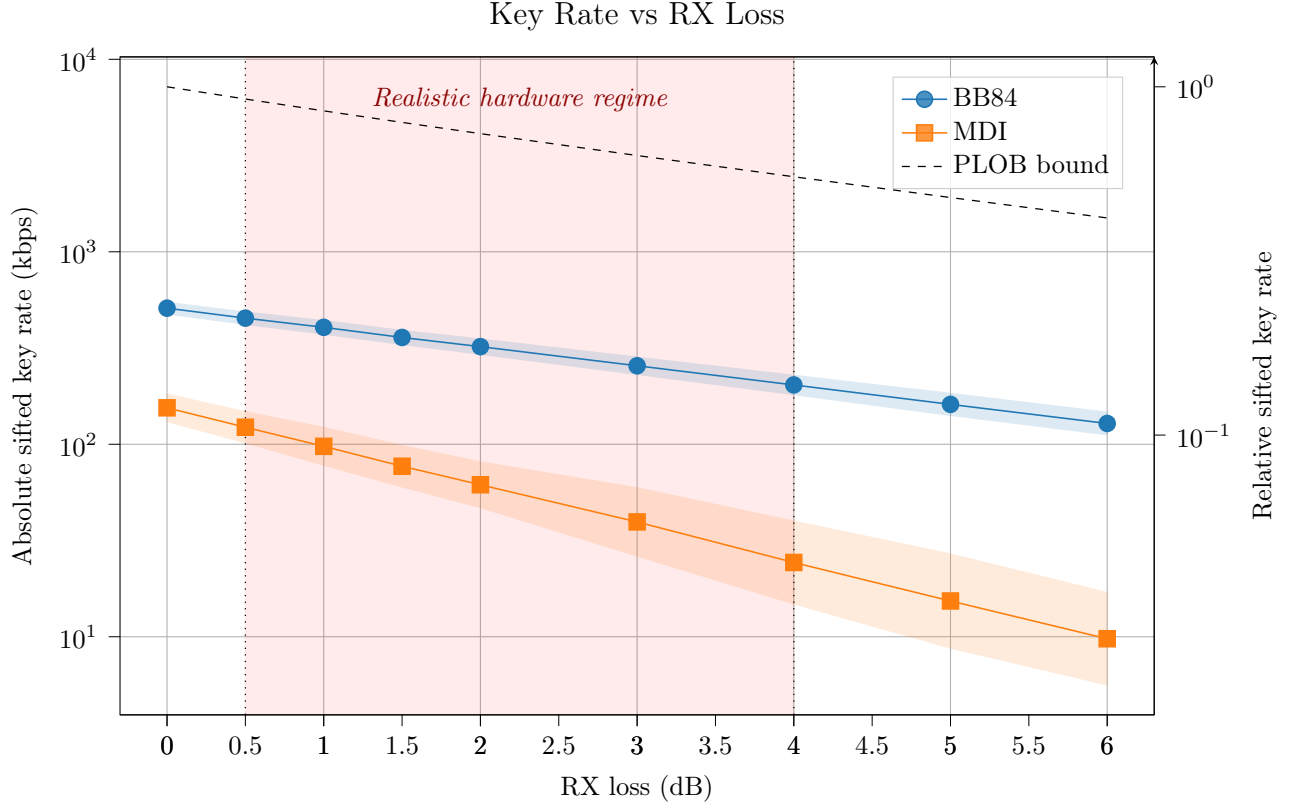


Figure 7: Absolute and relative sifted key rate versus receiver node/connector loss, at fixed 20 km separation and 0.18 dB/km attenuation. The PLOB bound is overlaid on the left axis; the shaded band marks the realistic hardware regime (0.5–4.0 dB).

Relay placement. Figure 8 sweeps Charlie’s position as a fraction of a fixed 20 km Alice–Bob distance.

MDI key rate peaks at the symmetric midpoint (61.7 kbps at 0.5) and falls by over a third towards either end (39.0 kbps at 0.1; 39.2 kbps at 0.9), confirming that Charlie sitting at the arm midpoint is rate-optimal. Moving Charlie off-centre widens the BB84/MDI ratio to $8.2\text{--}8.3\times$ at the sweep extremes, close to a 60% increase over the midpoint ratio of $5.2\times$: a clean illustration that relay geometry, and not only raw hardware quality, materially affects the rate trade-off of choosing MDI over BB84.

Sample spread. A further pattern, visible in the shaded bands of every sweep above

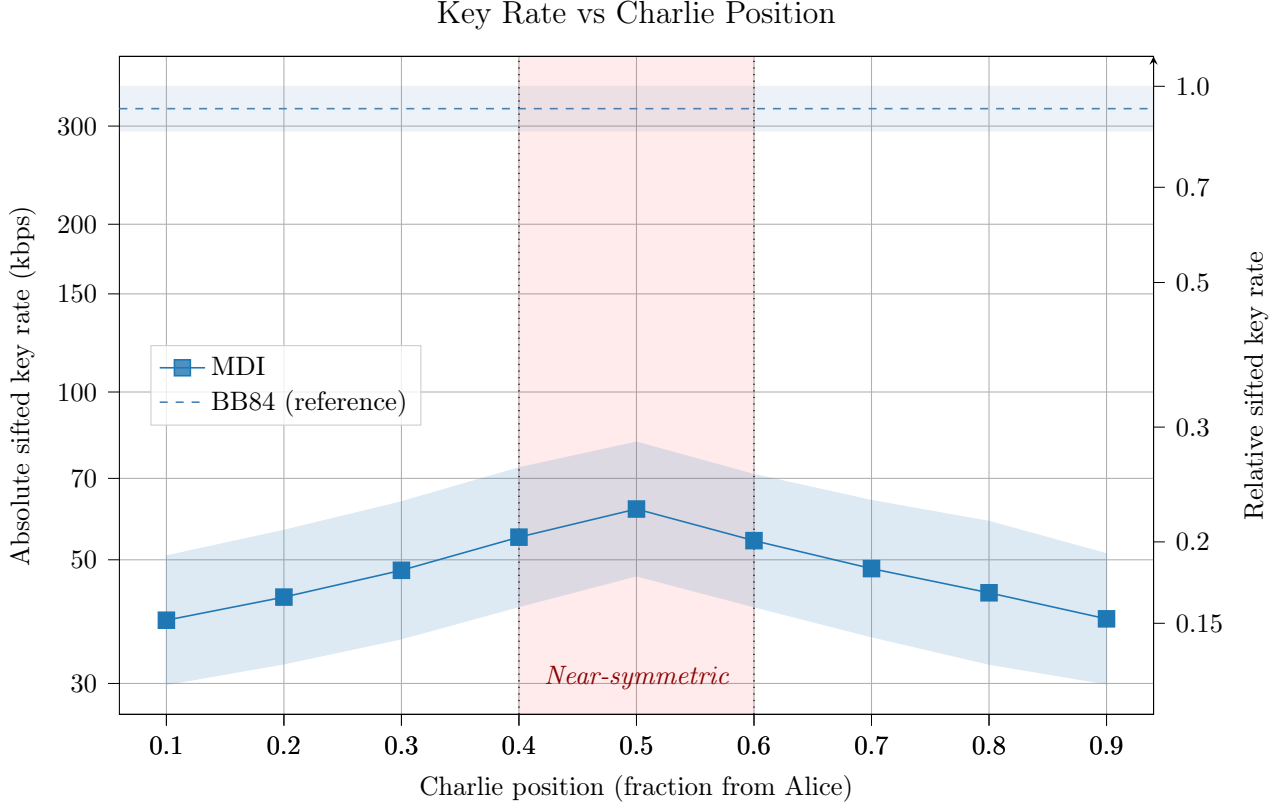


Figure 8: MDI key rate versus Charlie’s position as a fraction of the fixed 20 km Alice–Bob distance, with BB84 plotted as a distance-independent reference. The shaded band marks the near-symmetric regime.

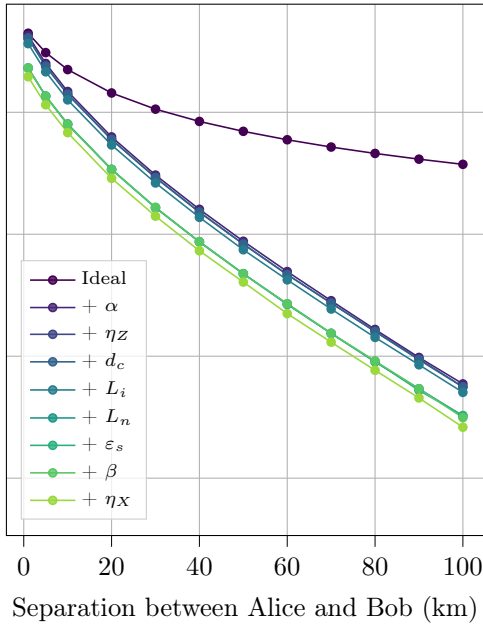
rather than in any single one, is that MDI’s sample spread is consistently wider than BB84’s relative to its own mean. Measuring each shaded band’s half-width as a fraction of its midpoint value and averaging across a sweep, MDI runs $1.7\text{--}3.3\times$ wider than BB84 at the same points: distance (0.71 vs. 0.41), receiver loss (0.65 vs. 0.20), detector efficiency (0.69 vs. 0.28), and Charlie position (0.54 vs. 0.18). This is not an artefact of any one parameter and has two compounding causes. First, MDI’s key generation is conditioned on a coincidence event at the BSM requiring both Alice’s and Bob’s photons to survive their respective arms, so an unlucky joint outcome on either arm alone is enough to suppress a trial’s yield. Second, because MDI’s absolute key rate is intrinsically lower than BB84’s at every operating point studied here, the same Monte Carlo budget yields fewer successful key-generation events per point, so data points carry proportionally more sampling noise by default. The practical consequence is that MDI’s key rate is more variable than BB84’s under otherwise identical conditions, a further, statistical, cost against which its device-independence guarantee must be weighed (Section 5.1).

Together, these four sweeps indicate that within the metropolitan and realistic hardware

regimes studied here, the BB84/MDI key rate gap is set primarily by a roughly constant structural factor of $\sim 5\times$ arising from the two-photon BSM requirement, and that per-arm hardware parameters – detector efficiency and receiver loss – are what is most likely to widen this gap substantially in practice. We test this conclusion directly with two further views of the same operating point: the cumulative hardware configurations of Table 2 each distance-swept, and a fixed-distance ranking of all nine parameters by relative impact; a further isolated per-parameter distance sweep connecting the two is reported in Appendix A.

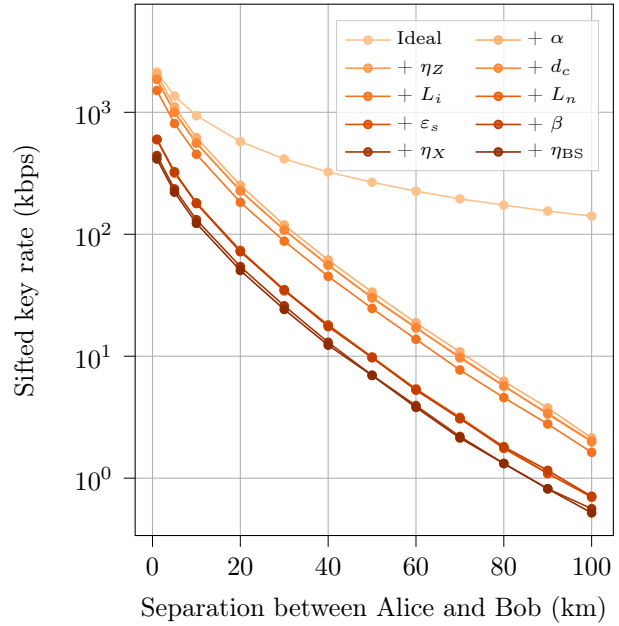
Cumulative hardware configurations. Figure 9 sweeps distance for each of the ten cumulative hardware configurations of Table 2 in turn.

BB84: Cumulative Modelling Layers



(a) BB84

MDI: Cumulative Modelling Layers



(b) MDI

Figure 9: Sifted key rate versus Alice–Bob separation under each cumulative hardware configuration of Table 2. Each configuration includes all impairments of the previous one plus the parameter named in its legend entry. Sample spread is omitted here for clarity (see full point-to-point sweeps for error bands).

Both protocols show the same qualitative shape: a large step when fibre attenuation is first added, followed by a sequence of comparatively small further steps as the remaining impairments accumulate. For BB84 the step from the ideal baseline to fibre attenuation alone is barely visible at 1 km – 4.44 Mbps falls by 3.9% to 4.27 Mbps – but dominates entirely by 100 km, with the same step reducing the rate by 98.4%. MDI shows an identical pattern: 3.4% at 1 km widening to 98.5% at 100 km. The fractional impact of fibre attenuation is itself a function of distance, negligible at short range and overwhelming at long range, consistent with it being the sole impairment whose loss probability scales with L directly (Section 3.1.2)

rather than acting as a fixed multiplicative factor. Against this backdrop, node loss stands out among the remaining impairments as the one further step large enough to see clearly even at short range: for BB84 adding node loss at 1 km induces a 36.8% key rate reduction; for MDI, a 60.4% reduction. Both figures are consistent with the sensitivity ranking below, and both closely match the isolated parameter impact reported in Appendix A.

Parameter sensitivity ranking. Figure 10 ranks all ten hardware parameters directly by their individual contribution to key-rate loss, at a fixed 25 km operating point (ideal key rate 1223.40 kbps for BB84, 481.63 kbps for MDI), with all other parameters held ideal. Detector efficiency contributes two bars, one for the SPAD (IDQ ID230, $\eta_d = 0.20$) and another for the SNSPD (IDQ ID281, $\eta_d = 0.90$).

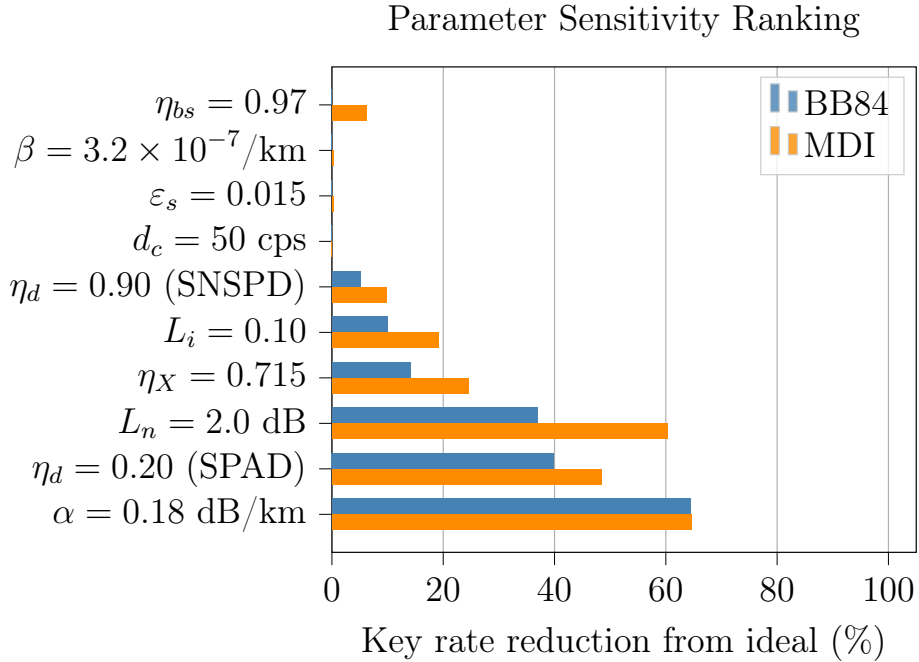


Figure 10: Percentage key-rate reduction from the ideal baseline at 25 km separation, for each parameter set individually to its realistic value (Table 2), ranked by impact on BB84 key rate.

Fibre attenuation dominates both protocols almost identically at this fixed distance (64.4% BB84, 64.7% MDI), expected for matching total fibre lengths. The SPAD detector-efficiency bar is the second-largest contributor for BB84 at a 39.9% reduction, and impedes MDI by 48.4%; SNSPDs, comparatively, produce 5.1% and 9.8% reductions, respectively. Other parameters reproduce the per-arm asymmetry established in Section 4.1: node loss reduces BB84 by 36.9% but is MDI’s second-largest contributor to performance reduction, with a 60.3% drop-off. Basis bias reduces BB84 by 14.1% versus 24.5% for MDI; TX insertion loss by 10.0% versus 19.2% – each roughly 1.6–2 $\times$ more damaging to MDI than to BB84. Beam-splitter efficiency, applicable to MDI alone, contributes a further 6.3%. Dark

count rate, source error rate, and fibre dephasing are negligible for both protocols at this operating point (under 1% in every case), matching the "noise-like parameters" grouping of Appendix A.

4.2 Network Performance

The user-count and relay-count sweeps of Section 3.3 are reported together here, and both characterise network-level key rate as a function of the same underlying variables, N and K .

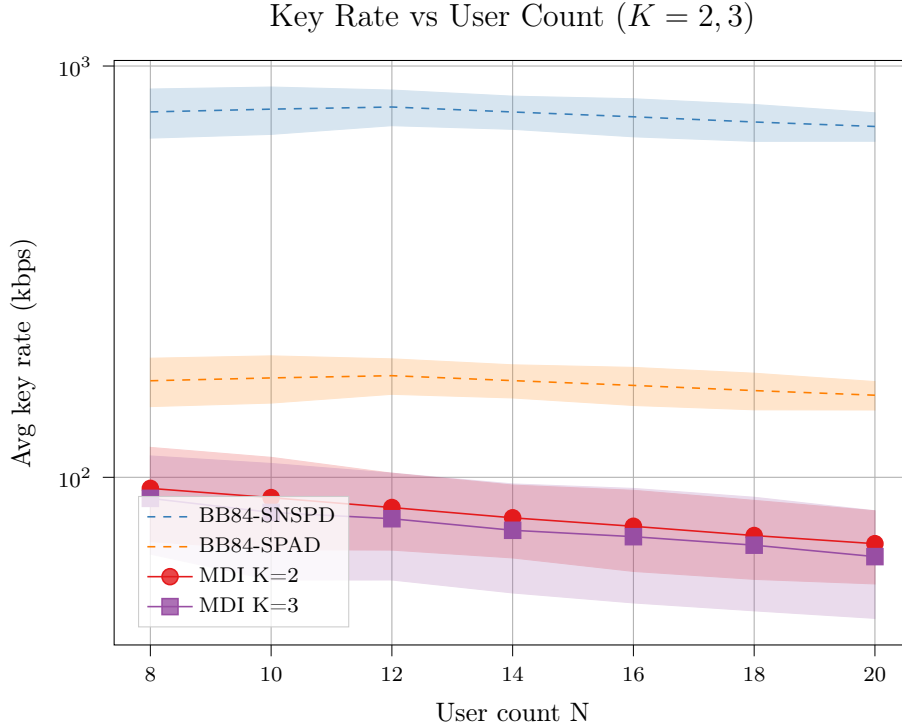


Figure 11: Mean key rate versus user count N under the organic-growth model of Section 3.3: $K = 2$ and $K = 3$ relay positions are placed once at $N = 8$ via the Weiszfeld iteration and held fixed as users join a fixed catchment one at a time in a 25×25 km service area, averaged over 5 independent seeds; shaded bands show the spread. BB84 is relay-independent and shown for both SNSPD and SPAD detectors.

Figure 11 shows MDI's mean key rate falling monotonically as N grows, from 94.0 to 69.0 kbps for $K = 2$, and 88.9 to 64.2 kbps for $K = 3$, both around a 27% decline. BB84 changes by under 8% over the same range (773.6 to 713.0 kbps SNSPD; 171.7 to 158.3 kbps SPAD), with individual seeds peaking anywhere from $N = 10$ to $N = 16$ rather than sharing a common shape. The asymmetry follows from what each protocol's rate depends on: BB84's point-to-point rate is set by each pair's own fixed separation (a noisier statistic for smaller N), invariant to how many other users share the network, whereas MDI's spoke distance to an unrelocated relay grows as new users land further from that catchment's original centre, directly eroding rate. $K = 2$'s mean sits consistently above $K = 3$'s at every N tested

(5–9%), though the effect is driven disproportionately by one of the five seeds tested with all others varying in sign, so the results don’t outright support a directional lean. What the sweep does establish robustly is that MDI, unlike BB84, is structurally sensitive to growing into infrastructure sized for a smaller network.

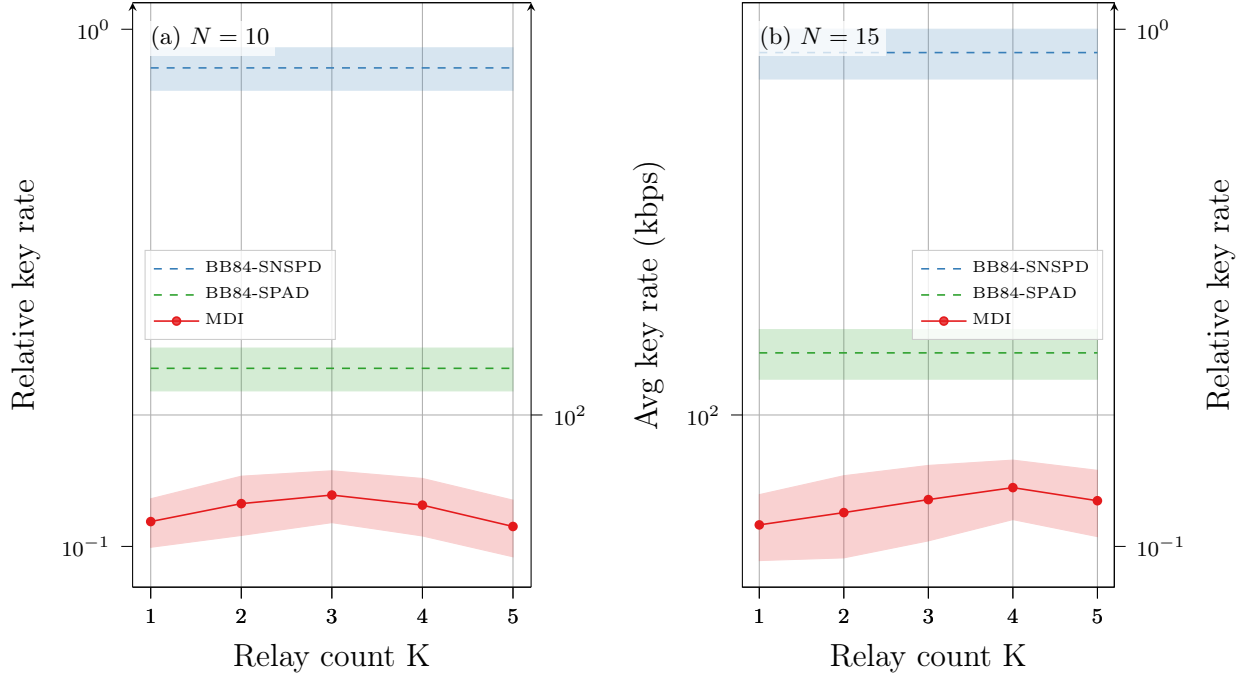


Figure 12: Mean key rate versus relay count K for $N = 10$ (left) and $N = 15$ (right) users in a 25×25 km service area, each averaged over 5 independent seeds; shaded bands show the spread across seeds. BB84 is relay-independent (dashed baselines) and indicated for both SNSPD and SPAD detectors. Both panels share the same y -axis scale on both axes to support direct visual comparison.

Figure 12’s relay sweep shows MDI’s mean key rate peaking at $K = 3$ for $N = 10$ (67.0 kbps, up 14.2% from $K = 1$, recovering 11.8% of the BB84 SNSPD baseline) and at $K = 4$ for $N = 15$ (69.5 kbps, up 20.5%, recovering 11.3%), falling off past the peak at both user counts (Figure 12). This non-monotonic shape reflects two effects of opposite sign: more relays shorten the mean user-to-relay spoke distance, improving the coincidence probability, but also fragment the network into smaller catchments, raising the chance a user pair must route via an inter-relay backbone hop. The location of the peaks could imply that the rate-optimal K scales like $\sqrt{N}$, discussed in more detail in Section 6.3, though the recovered fraction of the BB84 baseline is practically unchanged. Table 4’s per-seed breakdown for $N = 15$ confirms the mean’s $K = 4$ peak is not an averaging artefact: four of five seeds peak in key rate there outright, and the fifth is within Monte Carlo noise of doing so.

Total fibre length, by contrast, is minimised around $K = 2$ – 3 at both user counts, not at the key-rate optimum (Table 4): the backbone term grows quadratically in K (Equation 6)

Seed	Metric	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 5$
87100	Key rate (kbps)	48.5	47.2	54.4	62.9	56.3
	Fibre length (km)	145.8	123.8	115.8	129.0	143.3
29252	Key rate (kbps)	64.8	74.5	80.7	78.9	76.3
	Fibre length (km)	121.0	87.0	88.1	105.7	115.8
44358	Key rate (kbps)	51.2	50.5	51.7	55.8	55.8
	Fibre length (km)	137.9	111.3	112.0	126.0	135.2
91457	Key rate (kbps)	51.2	56.9	61.4	66.4	57.1
	Fibre length (km)	134.5	104.7	101.9	118.6	131.0
39964	Key rate (kbps)	72.7	77.6	79.2	83.6	79.9
	Fibre length (km)	112.3	84.6	89.0	98.2	107.3
Mean	Key rate (kbps)	57.7	61.3	65.5	69.5	65.1
	Fibre length (km)	130.3	102.3	101.3	115.5	126.5

Table 4: MDI mean key rate and total fibre length (Equation 6, straight-line Euclidean distance without stochastic tortuosity) versus relay count K , per-seed, for $N = 15$. Bold marks each row’s optimum.

while the spoke term can only fall so far, so past that point each further relay costs more backbone fibre than it saves on spokes, even as it continues to help key rate by shortening the worst-case spoke distance. The fibre-cost-optimal and key-rate-optimal relay counts are therefore different operating points: pushing from the fibre-optimal K to the key-rate-optimal one costs roughly 13–14% more fibre at $N = 15$.

A cheaper SPAD detector ($\eta_d = 0.20$) does not close the gap with SNSPD-MDI at either user count: the SPAD-BB84 baseline (126.3 kbps at $N = 10$, 136.5 kbps at $N = 15$) sits roughly $1.9\text{--}2.0\times$ above even the SNSPD-MDI peak, with no overlap between the two distributions at either N . Any cost comparison between the two protocols (Section 4.3) must therefore be read as a comparison at vastly unequal key rates, not as a search for a SNSPD-MDI deployment matching SPAD-BB84’s throughput.

4.3 Cost

Using the component counts of Table 3 (Section 3.2), the total CapEx for each protocol is

$$C_{\text{BB84}} = N c_s + 2N c_d + \frac{N(N-1)}{2} \bar{d}_b c_f, \quad (8)$$

$$C_{\text{MDI}} = N c_s + 4K c_d + \left(N \bar{d}_u + \frac{K(K-1)}{2} \bar{d}_r \right) c_f, \quad (9)$$

where $\bar{d}_b$ is the mean user-pair distance in the BB84 mesh, $\bar{d}_u$ the mean user-to-relay distance, and $\bar{d}_r$ the mean relay-to-relay backbone link length. Passive relay hardware (HOM beam splitters, polarising beam splitters, optical switches) contributes small additional fixed cost

per relay and is omitted here for simplicity; a deployed metropolitan MDI-QKD network using exactly this apparatus is reported in [19].

The fibre term dominates the cost difference at large N , since it scales as $\mathcal{O}(N^2)$ for BB84 against $\mathcal{O}(N)$ for MDI at fixed K . Thus, a crossover user count N^* past which MDI is cheaper always exists. Setting $C_{\text{BB84}} = C_{\text{MDI}}$ and solving for N gives N^* as a function of the ratio c_d/c_f , relay count K , and network geometry. Evaluating N^* explicitly requires relations between the distance term $\bar{d}_b$ in Equation 8 and the terms $\bar{d}_u$ and $\bar{d}_r$ in Equation 9.

Neither is a free parameter in practice: both follow from standard results in geometric probability, assuming relay placement is roughly uniformly across the service area (Section 3.1.4). For m points scattered independently and uniformly over a square of side L , the mean pairwise distance converges to

$$\overline{|PQ|} = \frac{2 + \sqrt{2} + 5 \operatorname{arcsinh}(1)}{15} L \approx 0.5214 L, \quad (10)$$

independent of m [49]. Setting $m = N$ is what already makes $\bar{d}_b$ itself independent of N in Equation 8; setting $m = K$ instead applies the same fact to the K relay positions, giving

$$\bar{d}_r \approx \bar{d}_b. \quad (11)$$

Spoke distance is a different quantity. K relays partitioning the service area into roughly equal-area catchments gives each catchment area $\approx L^2/K$, i.e. linear (side) dimension $\ell \approx L/\sqrt{K}$; each user's assigned relay sits near that catchment's geometric median, close to its centre. The mean distance from a point drawn uniformly at random within a square of side ℓ to that square's own centre is a second, distinct geometric-probability constant [49]:

$$\overline{|PO|} = \frac{\sqrt{2} + \ln(1 + \sqrt{2})}{6} \ell \approx 0.3826 \ell. \quad (12)$$

Substituting $\ell \approx L/\sqrt{K}$ gives $\bar{d}_u \approx 0.3826 L/\sqrt{K}$; eliminating the shared, unknown L using Equation 10 in its place yields the working approximation used below:

$$\bar{d}_u \approx \frac{0.3826}{0.5214} \cdot \frac{\bar{d}_{\text{BB84}}}{\sqrt{K}} \approx 0.73 \frac{\bar{d}_b}{\sqrt{K}}. \quad (13)$$

Both approximations are taken in the regime $N \gg K$: each relay serving many users, characteristic of realistic large-scale MDI-QKD deployment. In this regime a catchment's users fill its area densely enough for the uniform, independent scattering assumed by Equations 10 and 12 to hold.

For the analysis that follows, we additionally assume multi-channel detector modules suitable for every node's needs. The IDQ ID281 SNSPD system's 16 detection channels evidences that just one module can be necessary for a single node's detector channel requirement. Under this assumption BB84's detector cost becomes $N c_d$ and MDI's becomes $K c_d$, assuming symmetric cost of detector systems. We write $d \equiv \bar{d}_b$, and c_F for $c_f d$, the average cost of a link in full-mesh.

Substituting Equations 11 and 13, together with the multi-channel detector assumption above, into Equations 8–9 gives a closed form for the cost difference:

$$\Delta C \equiv C_{\text{BB84}} - C_{\text{MDI}} \approx c_d(N - K) + c_F \left[\frac{N(N-1)}{2} - 0.73 \frac{N}{\sqrt{K}} - \frac{K(K-1)}{2} \right]. \quad (14)$$

Expanding as a polynomial in N ,

$$\Delta C \approx \frac{1}{2} c_F N^2 + \left(c_d - c_F \left(\frac{1}{2} + \frac{0.73}{\sqrt{K}} \right) \right) N - \left(c_d K + \frac{1}{2} c_F K(K-1) \right), \quad (15)$$

with no further approximation. The linear coefficient's bracketed factor $\frac{1}{2} + \frac{0.73}{\sqrt{K}}$ decreases monotonically in K towards $\frac{1}{2}$ as $K \rightarrow \infty$; the coefficient itself therefore exceeds $c_d - c_F$ for all $K \geq 3$, and falls only marginally below it at $K = 2$.

Equation 15 is quadratic in N and solves in closed form. Writing $\rho \equiv c_d/c_F$ for the dimensionless ratio of single detector-module cost to average full-mesh link cost,

$$N^*(\rho, K) = - \left(\rho - \frac{1}{2} - \frac{0.73}{\sqrt{K}} \right) + \sqrt{\left(\rho - \frac{1}{2} - \frac{0.73}{\sqrt{K}} \right)^2 + 2\rho K + K(K-1)}. \quad (16)$$

$N^*(\rho, K)$ is monotonically decreasing in ρ for K fixed: differentiating Equation 16 and simplifying, so its value at $\rho = 0$ is a hard ceiling:

$$N^*(0, K) = \left(\frac{1}{2} + \frac{0.73}{\sqrt{K}} \right) + \sqrt{\left(\frac{1}{2} + \frac{0.73}{\sqrt{K}} \right)^2 + K(K-1)}. \quad (17)$$

For $K \geq 3$, the first term is less than 1, and the square root is $\leq K$. N^* evaluates to 2.76 at $K = 2$, and so for each fixed $K \geq 2$, $N^* \leq K + 1^1$. Aside from the fact that the simulation data points to MDI key rate peaking for $K = \sqrt{N}$, economic deployments of MDI-QKD would target far fewer relays than users (i.e. $K \ll N$). Thus, for every realistic (N, K) combination, $N > N^*(\rho, K)$ for *every* $\rho \geq 0$, so **MDI is the cheaper protocol to deploy at every operating point, regardless of detector price** conditioned only on symmetric price for detector systems across protocols.

¹ $N^*(0, 1) = 2.46$, the *smallest* crossover value at any relay count. Even so, only $N = 3$ users are needed for MDI to be strictly cheaper than BB84 at $K = 1$, regardless of detector price.

If $K = N$, then $\Delta C(K) = -0.73 c_F \sqrt{K}$, negative for every $K > 0$ and independent of c_d (the detector term vanishes identically). BB84 thus remains the cheaper protocol if all users are also relays, regardless of detector price, and a crossing never occurs. $K = N$, however, is precisely the deployment no operator would choose: a relay serving a single user shares no fibre and provides none of the aggregation MDI's hub topology exists to offer, and so the result above holds across the entire range of sensible deployments.

The K floor is exactly attained in the limit. Writing $a \equiv \rho - \frac{1}{2} - \frac{0.73}{\sqrt{K}}$, the identity $\sqrt{a^2 + c} - a = c/(\sqrt{a^2 + c} + a)$ applied to Equation 16 (with $c = 2\rho K + K(K-1)$) gives

$$N^*(\rho, K) = \frac{2\rho K + K(K-1)}{\sqrt{a^2 + 2\rho K + K(K-1)} + a}. \quad (18)$$

The numerator grows as $2K\rho$ and the denominator as 2ρ , so $N^*(\rho, K) \rightarrow K$ as $\rho \rightarrow \infty$, with $N^*(\rho, K) - K \sim 0.73\sqrt{K}/\rho$ at next order. Combined with the $\rho = 0$ ceiling above, N^* is confined to the open interval $(K, K+1)$: less than one user's width, regardless of how cheap or expensive detectors become.

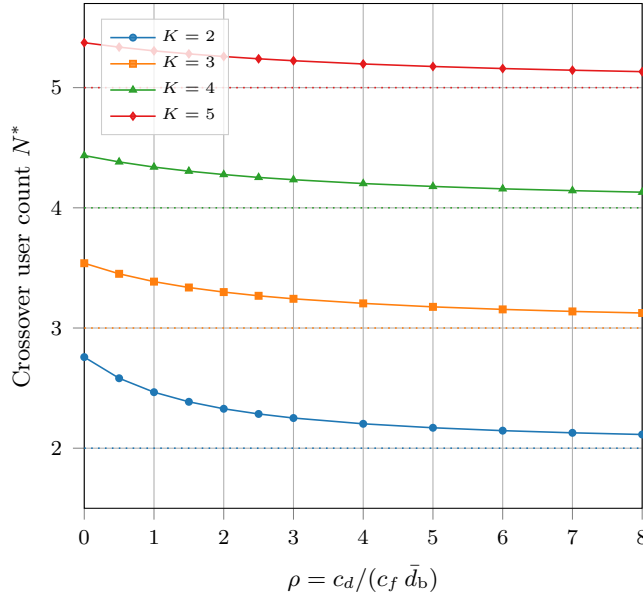


Figure 13: Crossover user count N^* against the dimensionless price ratio $\rho = c_d / (c_f \bar{d}_b)$, from the closed-form solution of Equation 16, for $K = 2-5$; dotted lines mark each curve's $N^* \rightarrow K$ asymptote, approached from above.

Figure 13 makes this visible directly: each curve falls from its $\rho = 0$ ceiling towards its floor (dotted) as $\rho \rightarrow \infty$.

Marginal cost of network growth. The comparison above treats N as fixed; a provider deciding whether to extend an existing network by one more user instead faces a marginal-cost question, fixing K . Differencing Equations 8–9 (with the multi-channel detector assumption)

between N and $N + 1$ gives the cost of the $(N+1)$ -th user under each protocol:

$$\Delta C_{\text{BB84}}^{\text{marg}}(N) = c_s + c_d + c_F N, \quad \Delta C_{\text{MDI}}^{\text{marg}}(N) = c_s + c_f \bar{d}_u \approx c_s + 0.73 \frac{c_F}{\sqrt{K}}. \quad (19)$$

BB84’s marginal cost grows linearly with the size of the network the new user joins, since it must connect to all existing ones. MDI’s marginal cost is instead constant in N : the new user needs only one spoke link, of mean length $\bar{d}_u$, to its nearest relay, and shares that relay’s already-provisioned detector module. The difference between the two,

$$\Delta C_{\text{BB84}}^{\text{marg}}(N) - \Delta C_{\text{MDI}}^{\text{marg}}(N) = c_d + c_F \left(N - \frac{0.73}{\sqrt{K}} \right), \quad (20)$$

is strictly positive for deployments with $N \geq 1$ and $K \geq 2$ regardless of c_d and c_f : **adding a user to an existing MDI network is cheaper than adding one to a BB84 mesh of the same size**, at every step, and this **does not require symmetric detector costs**. This is the mechanism behind the tight N^* ceiling derived above: BB84 is only ever cheaper because of a smaller *initial* outlay, and every user added costs strictly more, so that initial head start is quickly lost. In this light N^* marks the point at which BB84’s one-off discount is fully repaid.

One caveat attaches to the argument above: custom efficiency and detection channel count devices do not yet exist, and we make the extra assumption that multi-channel detectors cost the same regardless of protocol requirement. BB84 has weaker requirements on each of these, seen in Figure 11 and Equation 8, to achieve MDI key rates – but as outlined initially, fibre scaling means that there will still always be a crossing point N^* for large enough networks.

4.4 Deployment Recommendations

Cost and key rate are not the only axes that differentiate the protocols in this work: MDI-QKD’s principal selling point is its security guarantee, not its throughput. It is useful to ask directly what a deployment requires the customer to trust: the photon source, the detectors, any relay hardware, and the people operating that hardware. This is what varies between architectures and what a customer’s threat model actually constrains. One component never varies: the quantum channel itself (the fibre, and by extension any eavesdropper on it) is assumed fully adversarial at every level – that is the premise QKD exists to satisfy. Customer trust requirements give a strict hierarchy, each assignable a set of protocols:

- **Level 0: Trust nothing.** Not the source, not the detectors, not any relay, not any operator. This is the ceiling fully Device-Independent QKD aims for, first proposed

with a security proof against collective attacks by [50] and recently demonstrated experimentally [51]; it is not implemented in this work.

- **Level 1: Trust in sources.** Detectors, relay hardware, and the relay operator are all permitted to be fully adversarial: Charlie’s Bell-state measurement discloses only the joint correlation between Alice’s and Bob’s photons (Section 5.1). The source is still trusted, though multi-photon emission and Trojan-horse side channels are residual vulnerabilities that decoy-state protocols [11, 31] address but the ideal sources used in this work do not model directly (Section 5.3). No substitute exists at this level: a customer whose threat model stops at Level 1 has no BB84 alternative.
- **Level 2: Trust in sources and detectors.** Adds detector trust to Level 1’s source trust, in exchange for removing the relay from the picture entirely: with no third party involved, there is nothing further to trust, but the link cannot extend beyond direct point-to-point range. Detector trust is precisely the assumption BB84 requires and MDI was invented to remove (Section 5.1), so this level is only appropriate when both endpoints already fall within a single security perimeter.
- **Level 3: Trust everything.** Adds trust in relay hardware *and* its operator: a compromised relay node, or a dishonest operator, recovers the full end-to-end key for any pair routed through it (Section 5.1). In exchange, range extends via relay hops at BB84’s rate and cost, without MDI’s two-photon coincidence penalty. Appropriate only when the relay is under the customer’s own physical and operational control, or is otherwise contractually and regulatorily accountable, as in deployed trusted-node metropolitan QKD networks such as the 46-node Hefei network [52].

Each level’s trust set strictly contains the one below it, so a customer’s threat model fixes a level first, and only *then* does the cost/rate trade-off apply across suitable architectures. A customer requiring Level 1 is not choosing between BB84 and MDI at all: MDI is the only option meeting the requirement.

Protocol selection follows from combining the simulation key rate results with the algebraic cost model. The relay sweep provides mean key rate at fixed topology; Equations 8–9 express cost as a function of the same variables and component costs.

For a new deployment with a minimum key rate service requirement, a provider would identify the suitable relay count(s). The cost of an MDI network can then be compared to the minimal cost of a suitable BB84 network, indicating a difference in value. The crossover user count provides a scale threshold.

5 Discussion

The results presented in Section 4 support a consistent picture of the trade-off space between BB84 and MDI-QKD. We discuss the key findings, their physical interpretation, their implications for protocol selection, and the limitations of the methodology.

5.1 The Security, Rate, and Cost Trade-off

MDI-QKD’s elimination of required trust in photon detectors comes at a key-rate penalty rooted in its physical architecture. Whereas BB84 requires one photon to traverse a link and survive detection, MDI needs one from each party to arrive at Charlie’s relay simultaneously and produce a coincidence event; component losses on each arm therefore compound multiplicatively, scaling channel transmittance ($\eta < 1$) quadratically. This is consistent with the point-to-point sweeps (Section 4.1): a roughly constant $\sim 5\times$ gap under channel-side sweeps (distance, attenuation) that widens past $25\times$ under detector efficiency and to $13\times$ under receiver loss. Framed this way, the headline understates the risk a real deployment runs: it holds only when both protocols are given matched-quality detectors. BB84 tolerates the cheap SPADs used in the network-level sweeps, while still outperforming MDI, simulated here with premium SNSPDs: SPAD-BB84 still beats SNSPD-MDI rates by $1.9\text{--}2.0\times$. The penalty derived from matched hardware is therefore only a floor on the value gap.

The penalty compounds at network scale. MDI users route to a relay before reaching their partner, an overhead absent from BB84’s direct mesh; holding relay infrastructure fixed while N grows costs MDI in performance (see the $\sim 27\%$ rate decline against BB84’s under 8% , Figure 11). Adding relays to counter this recovers only a small portion ($9.4\text{--}11.3\%$ in Figure 12) of the gap before the recovery itself reverses. This is a property of the fixed infrastructure model used here, not of MDI – a provider willing to periodically re-site relays could plausibly arrest the decline (Section 6.3). MDI’s key rate is also intrinsically noisier than BB84’s under identical sampling budgets: shaded-band half-widths run $1.7\text{--}3.3\times$ wider relative to their own mean across every point-to-point sweep tested (Section 4.1), because a BSM coincidence event depends on two photons surviving and because MDI’s architectural increased loss yields fewer successful trials. An operator provisioning for a minimum rate, rather than an expected one, pays for this variance in extra headroom capacity – a cost of device-independence that a mean-rate comparison does not surface.

Set against all of this, Section 4.3’s cost model delivers the headline economic result: under a shared detector price, MDI is cheaper to deploy at every realistic user-relay combination, and its marginal cost of serving one more user is constant where BB84’s grows linearly

(Equation 19). At face value, this could read as MDI's economic case standing on its own, independent of its security motivation, though that reading does not survive contact with the rate results above. Cost and key rate answer different questions – what a network costs to build, against what it can deliver once built – and are not substitutes for one another: a network priced below its BB84 alternative is worthless to a customer whose throughput requirement it cannot meet. The cost result is also considerably more fragile than its closed form suggests: it rests on pricing BB84 and MDI detector modules identically and on CapEx alone, excluding the cryogenic operating cost an SNSPD-equipped MDI relay incurs and a SPAD-equipped BB84 endpoint does not (Section 5.2). Relax either assumption in MDI's disfavour and "always cheaper" is no longer guaranteed to hold at every (N, K) ; the true economic case for MDI is conditional in a way Equation 16's derivation does not expose.

The picture that emerges is asymmetric rather than balanced. BB84 wins unconditionally on rate, at every distance and hardware configuration tested, and where it loses on cost, it loses permanently: its fibre term keeps growing relative to MDI's (Section 4.3), so BB84's disadvantage compounds. MDI's claimed cost advantage, by contrast, holds only under assumptions – symmetric detector pricing, no operating expenditure, static relay infrastructure – that this work does not independently verify. What does not depend on any of these assumptions is MDI's security guarantee: unlike its cost advantage, it is unconditional, and a customer whose threat model demands it has no BB84 substitute at any price or rate (Section 4.4's Level 1 trust boundary). Furthermore, integrating QKD into existing infrastructure favours MDI greatly – irrespective of the price of hardware (Equation 20). The trade-off this work quantifies is therefore less an even contest between "cheap-but-slow" and "fast-but-expensive" than the results might first suggest: MDI's case rests almost entirely on that security guarantee, with its apparent cost advantage better read as a fragile bonus than as an independent reason to prefer it.

5.2 Cost Model Limitations

The cost model is a first-order CapEx-only algebraic estimate. Several factors are not captured.

Component price data. Manufacturer prices for quantum dot photon sources and SNSPD detectors are not publicly available; commercial quotes are required for direct quantitative comparison. This matters less for the crossover conclusion than it might: N^* is provably confined to $(K, K+1)$ for every price ratio $\rho = c_d/c_F$ (Section 4.3), so no realistic uncertainty in c_s , c_d , or c_f can move the qualitative result that MDI becomes the cheaper protocol. Price uncertainty instead matters for magnitude, not direction: how large ΔC grows beyond N^* , and any absolute cost figure the model produces, scale directly with component pricing.

Symmetric detector pricing. The model prices both protocols’ detector modules at a single shared c_d (Section 4.3). Real prices could break this symmetry either way: MDI’s BSM demands tight timing and efficiency matching, plausibly commanding a premium, while the low-specification SPADs BB84 tolerates (Section 4.2) could just as easily give it its own rate and cost advantage. Every cost result should therefore be read at a shared, not necessarily optimally faithful, detector price; Section 6.3 proposes relaxing this directly.

Operational expenditure. Cryogenic cooling for SNSPD detectors and QD sources (order 2–4 K) is an ongoing energy cost the CapEx-only model omits entirely, and plausibly a non-trivial one for the N QD-equipped users and K SNSPD-equipped MDI relay stations at scale; Section 6.3 proposes incorporating it via the energetic analysis methodology of [53].

BB84 fibre scaling at large N . The BB84 direct mesh assumes one independent dark-fibre run per user pair. In practice, wavelength-division multiplexing (WDM) or trustless switched architectures would share fibre infrastructure at high N , substantially reducing BB84 fibre cost. The $\mathcal{O}(N^2)$ fibre scaling should therefore be interpreted as an upper bound applicable to passive, unmanaged mesh deployments; it is physically exact for that specific case.

5.3 Simulation Model Limitations

Source brightness and insertion loss decoupling. The Quandela Prometheus single-photon source [39] used as the reference product operates at 975 nm and achieves a minimum fibred brightness of 40%, with frequency conversion required to reach the 1550 nm telecom band. In our simulation, source brightness is absorbed into the effective source repetition rate, and the TX loss parameter models separate optical path losses independently. Frequency conversion efficiency, additional conversion-stage noise, and the explicit coupling between source brightness and photon rate are not modelled. This is a deliberate simplification; a full model would propagate each loss stage independently and account for Raman background photons introduced by conversion.

Ideal single-photon sources. All simulations assume one photon per pulse with no multi-photon component. WCP laser sources are overwhelmingly more practical and affordable than QD sources, and their multi-photon components require decoy-state analysis [31] to extract secure key rate. The ideal source assumption gives an upper bound on key rate and on source cost; both should be re-evaluated for WCP deployments.

Sifted, not secure, key rate. Every key rate reported here, including R in Equation 4, is sifted, not secure – matched-basis raw key length over simulation time (Section 3.1). No error-correction leakage term (Lütkenhaus [54]; GLLP [55]) or privacy amplification is subtracted, nor is a numerical bound such as Tan et al.’s [56] applied. Since the leakage term is QBER-dependent and QBER differs between protocols, one of these corrections is a

prerequisite before the gap reported throughout Section 4 can be read as secure, rather than sifted, throughput.

Photon-count normalisation. MDI’s two sources each prepare $n = 512$ photons per trial so the summed count ($2 \times 512 = 1024$) matches BB84’s $n = 1024$ (Section 3.1), but this halves MDI’s per-source pulse-train window (51.2 vs. 102.4 μs at $f = 10^7$ Hz). The fixed per-trial propagation and reconciliation overhead in $T_{\text{end}} - T_{\text{start}}$ (Equation 4) is independent of n , so it consumes a larger fraction of MDI’s shorter window – an artefact that depresses MDI’s rate for reasons unrelated to the protocols’ physical difference. Because this bias inflates MDI’s apparent disadvantage rather than creating it, it does not threaten the qualitative conclusion, but the reported BB84/MDI gap should be treated as an upper bound.

Asymptotic key rates. Finite-key corrections [57] become significant at low photon count rates and short block lengths, reducing the extractable key compared to the asymptotic rate reported here. At long distances where photon detection events are sparse, the finite-key penalty is expected to be most severe for MDI-QKD, potentially widening the gap with BB84 beyond what the asymptotic analysis suggests.

Relay placement. Relay positions are set by the generalised Weiszfeld iteration (Equation 7), which minimises total fibre length – but minimising fibre length is not the same as optimising network key rate or cost, and cluster assignment itself is only locally optimal (Section 6.3).

Topology sensitivity. The user-count sweep (Section 4.2) illustrates a broader point about every relay-placement and growth result in this work: outcomes are highly sensitive to the specific topology drawn, not just to N and K in aggregate. In Figure 11, the mean key-rate advantage of $K = 2$ is driven disproportionately by a single seed; the remaining four show variable preference to the different relay counts. Averaging over synthetic topologies, as this work does throughout, is a necessary simplification for a general-purpose comparison, but a real deployment sees exactly one topology. A provider sizing relay infrastructure should design around its own known or forecast user requirements first and foremost, with general trends informed by similar analysis to that presented here. Appendix C’s BT case study illustrates this with one real deployment; extending the same real-geography methodology to other operators’ published site layouts (Section 6.3) would show how much of that result is specific to BT’s topology versus general to metropolitan geography.

Receiver node loss. The model applies a single lumped $L_n = 2.0$ dB to both protocols’ receivers, though in reality BB84 (EOM, PBS) and MDI (HOM beam splitter, PBS, and, for cross-cluster routing, an optical switch) use different, separately-priced components with different insertion losses. The resulting bias is not negligible: Figure 10 shows node loss alone accounts for a 36.9% key-rate reduction for BB84 against 60.3% for MDI at 25 km, the single

largest per-arm contributor to the gap identified in this work, and MDI’s extra relay-side components mean a proper breakdown would plausibly alter this margin.

NetSquid computational cost. The choice to use NetSquid comes with the binding constraint of great computational cost to achieve the attained level of statistical significance. Appendix D quantifies this in full, along with proposed mitigations and what a commercial-scale successor would need.

Metropolitan network scope. The core comparison models a 25×25 km metropolitan grid, where both protocols reach useful key rates without range extension. Beyond that scale, links become loss-limited and two range-extending architectures become relevant. Trusted-node relaying restores range at the cost of end-to-end security, since a physically compromised relay exposes the key (Section 5.1). Quantum repeaters extend range via entanglement swapping and distillation without that trust assumption [58, 59], at the cost of quantum memories and mid-circuit measurement not yet proven at scale. Neither is generally modelled here, but the case for needing one is clear from our results: fibre attenuation is already the dominant term by 100 km for both protocols (Figure 9), precisely the loss regime a repeaterless link or quantum-internet-scale network [60] cannot operate in. Trusted-node relaying *is* implemented (TBB84, Section 3.1.4) and its rate and cost quantified in Appendix C.

5.4 Implications for Protocol Selection

The results support a protocol-selection framework based on two axes: security requirement and scale. Below the crossover user count N^* (Section 4.3), BB84’s direct mesh is both cheaper and faster than MDI: its bipartite key rate exceeds MDI’s by a roughly constant $\sim 5\times$ under channel-limited conditions (Section 5.1), and its fibre cost undercuts MDI’s despite the steeper $\mathcal{O}(N^2)$ asymptotic scaling. But N^* itself sits only just above the relay count K (Section 4.3), regardless of detector price, so this regime is narrow: for realistic (N, K) combinations, N exceeds N^* by default, and MDI’s marginal cost of serving additional users is unconditionally lower than BB84’s (Equation 19). The relay-count sweep of Section 4.2 shows K itself has a rate-optimal value distinct from its fibre-cost-optimal value (Table 4), so a provider optimising for key rate is pushed towards exactly the regime that keeps N^* low. Where the deployment scale exceeds N^* , MDI is simultaneously the cheaper choice and the slower one: choosing it on cost grounds is not a free win, since the same operating point still carries the rate and rate-variance penalty quantified in Section 5.1. The choice between BB84 and MDI is therefore a security question in the end, not a cost one: MDI eliminates detector side-channel vulnerabilities at the cost of a lower key rate, while BB84 delivers higher throughput within the trust assumptions of its detector model, and cost only ever determines which option is cheaper to buy, not which is better to have.

6 Conclusion

6.1 Summary of Contributions

This work has built a complete simulation and cost modelling infrastructure for the comparative analysis of BB84 and MDI-QKD at metropolitan scale. On the simulation side, both protocols are implemented within the NetSquid discrete-event simulation engine under a ten-configuration cumulative hardware model that progressively introduces physical imperfections from an idealised baseline to a fully realistic system. Point-to-point key rates are obtained via Monte Carlo simulation across a parameter space covering fibre loss, detector efficiency, dark count rate, TX insertion loss, receiver node loss, source error rate, fibre dephasing, detector basis bias, and BSM beam-splitter efficiency. The simulation infrastructure parallelises across CPU cores and persists all results to a structured SQLite database.

At the network level, both protocols are implemented as multi-user simulators: a direct BB84 mesh and an MDI-QKD network with relay BSM stations. Relay positions are placed by a generalised Weiszfeld iteration that minimises total fibre length, and all network experiments are repeated across multiple independent seeds to characterise topology variability. A symbolic cost model expresses total CapEx in terms of free parameters c_s (photon source), c_d (single-photon detector), and c_f (fibre per km), with component counts determined by topology. The model focuses on the structural scaling difference between BB84 and MDI and derives the crossover user count N^* analytically.

Beyond the BB84-versus-MDI comparison itself, this infrastructure was built to generalise. The methods applied to analysis of performance and cost, and the SQLite database (Appendix E) are not specific to any protocol. As such, one of this work’s principal contributions is its early-stage groundwork toward a general-purpose NetSquid-powered QKD network simulator, with engineering bottlenecks discussed in Appendix D.

6.2 Findings

In the bipartite setting, MDI-QKD’s key rate is structurally lower than BB84’s, a direct consequence of its two-photon BSM coincidence requirement (Section 4.1): roughly a constant $5\times$ lower under channel-limited conditions such as distance and fibre attenuation, since both protocols share the same link loss. The gap widens past $25\times$ wherever a per-arm hardware parameter degrades, since such parameters act quadratically on MDI’s coincidence probability but only linearly on BB84’s single-photon detection (Figure 10). Detector efficiency and receiver node loss are the dominant *tunable* hardware parameters limiting key rate at

short and medium range, while fibre attenuation dominates at long range for both protocols regardless of hardware choice, swallowing over 98% of key rate by 100 km (Figure 9).

This rate gap widens further at the network level through cross-cluster relay-routing detours (Section 5.1). Adding relays helps, but recovers only around a tenth of the BB84 baseline at best, and the recovery is non-monotonic: beyond some point dependent on user count, extra relays fragment the network into smaller catchments faster than they shorten user-to-relay distances (Table 4). The relay count that maximises key rate is also not the one that minimises deployment fibre, so a provider optimising purely for cost and one optimising purely for key rate reach different relay-count decisions even before component prices enter the picture. MDI is also the more growth-sensitive protocol: with relay infrastructure sized once and not revisited, its mean rate falls $\sim 27\%$ as N increases from 8 to 20, versus a reduction of under 8% for BB84 (Figure 11).

The symbolic cost model predicts that MDI becomes cost-competitive above a crossover user count, driven by its linear rather than quadratic fibre scaling with user count (Section 4.3). This crossover is reported as a function of relative detector and fibre cost and network geometry rather than as a single number. Under the assumptions used to derive it, MDI’s marginal cost of serving each additional user is unconditionally lower than BB84’s, and MDI is already the cheaper protocol at every network size simulated in this work.

6.3 Future Work

Several extensions are of immediate interest.

Finite-key corrections. Building on the asymptotic-rate limitation noted in Section 5.3, both the finite-key correction [57] and the secure-key-rate leakage term depend on QBER, which this work reports only in aggregate; decomposing it by contributing noise source (dark counts, dephasing, source error, basis bias) at each operating point would identify which mechanism to target first when tightening either correction.

Weak coherent pulse sources. Replacing ideal QD sources with WCP laser sources modelled via the decoy-state technique [31] would give rates more directly comparable to published experimental deployments [15, 19, 20] and would substantially reduce assumed source cost.

Asymmetric detector pricing. Section 5.2 noted the shared c_d assumption; replacing it with separate c_d^{BB84} and c_d^{MDI} terms and re-deriving N^* and the marginal-cost comparison (Equations 16 and 19) as functions of their ratio would give a more accurate value assessment than the current model.

Cost per unit key rate. Cost and key rate are compared on separate axes throughout: Section 4.3 compares total deployment cost at a given (N, K) , while Sections 4.1 and 4.2 compare key rate at a given hardware configuration, and the two are deliberately not com-

binned. A price-per-bit metric, $C(N, K)/\bar{R}(N, K)$ in \$/kbps would instead give one directly comparable deployment-efficiency number, incorporating the $\sim 5\text{--}25\times$ bipartite key-rate gap (Figure 6) into the cost comparison itself rather than treating it as a caveat alongside it.

Cost-rate bi-objective optimisation. A single \$/kbps ratio (above) collapses cost and rate into one number; keeping them as separate objectives asks the more operationally direct question: for a fixed budget $C_{\max}$, which $(K, N, \text{topology})$ maximises mean key rate, and for a fixed minimum rate $R_{\min}$, which minimises cost? Sweeping K and topology seed at fixed N against each configuration’s simulated key rate and the cost model’s fibre-length input would let the divergence between the fibre-cost-optimal and rate-optimal K already observed at $N = 15$ (Table 4) be characterised explicitly across the full trade-off curve, rather than read off two separate single-point optima.

Optimal relay placement. The generalised Weiszfeld placement used throughout this work (Section 5.3) minimises fibre length only. ILP-based placement [25], solving assignment and placement jointly, would instead optimise directly on network key rate and deployment cost, providing tighter lower bounds on minimum MDI deployment cost than either the Weiszfeld or k -means baselines compared in Appendix B.

Brownfield relay expansion. The growth-model experiments of Section 3.3 and Appendix B hold the relay count K fixed while users grow, modelling a provider serving unplanned demand on existing infrastructure. The relay count sweep of Section 3.3, by contrast, assigns all relays from scratch at each K , modelling deployment of a new network rather than expansion: a real operator adding capacity cannot costlessly relocate relays. Extending $K \rightarrow K + 1$ realistically would instead fix the existing relay positions and solve for where a single new relay should go, and which users it should serve.

Relay count sweep at further user counts. The relay count sweep of Section 4.2 showed MDI key rate peaking at $K = 4$ for $N = 15$ (Figure 12b, Table 4) and at $K = 3$ for $N = 10$ (Figure 12a), consistent with the optimal relay count scaling roughly as $\sqrt{N}$, though this rests on only two data points. More data points and a rigorous analytical model are needed to confirm this hypothesis.

Real network topologies. Appendix C’s BT case study already extends this work’s cost and key-rate comparison, and a cost-per-key-rate value assessment, to one real site layout; applying the same treatment to other operators’ published metropolitan deployments [15, 19, 20] would show how much of that comparison is BT-specific versus general to metropolitan geography, alongside the vendor-quoted, rather than illustrative, pricing the appendix itself flags as needed for a load-bearing cost figure.

Operational expenditure. Incorporating cryogenic and other OpEx, using the energetic analysis methodology of [53], would enable full lifetime cost comparisons and may substan-

tially alter the economic case for MDI at high relay counts.

Quantum repeater modelling. Trusted-node relaying is already implemented and costed (TBB84, Appendix C), but quantum repeaters – extending range via entanglement swapping and distillation without a trust assumption [58, 59] – are not modelled here. Simulating a repeater chain within the existing NetSquid infrastructure, whose extensibility to temporally correlated dynamics Appendix D discusses, would complete the comparison across the full distance regime beyond metropolitan scale.

Traffic modelling. Network key rates are the aggregate of all contained bipartite links. Letting a simulation of a network run live would relax the geometric-upper-bound assumption noted in Section 3.1.4 and give a more realistic picture of achievable, rather than ceiling, QKD behaviour. Every reported network rate here assumes all pairs draw key simultaneously at full channel capacity, with no contention, dead time, or scheduling overhead.

General-purpose QKD network simulator. The simulator architecture and cost modelling built here are protocol-agnostic by construction (Section 6). Generalising them into a framework – e.g. adding extra protocols and support for arbitrary deployment configurations – would extend this work into reusable infrastructure for QKD deployment planning similar to s-3 [27] or OMNeT++ [28] in classical network simulation.

A Additional Point-to-Point Parameter Sweeps

Section 4.1 discussed four of the eleven point-to-point parameter sweeps produced for this work: distance, detector efficiency, receiver loss, and Charlie’s relay position. Of the remaining seven, three – dark count rate, fibre dephasing rate, and source error rate – are omitted from Figure 14 below and quantified directly in the *Noise-like parameters* discussion instead, since each varies key rate by well under 3% across its realistic hardware range. Figure 14 reproduces the remaining four; the discussion below repeatedly compares these against detector efficiency (Figure 6) and receiver loss (Figure 7) from Section 4.1. In every panel, only the swept parameter deviates from its layer-9 default in Table 2; all other parameters are held at that default operating point (20 km Alice–Bob separation, 0.18 dB/km fibre attenuation, $\eta_d = 0.9$ detector efficiency, 50 cps dark count rate, $L_i = 0.1$ TX insertion loss, 2.0 dB node loss, $\varepsilon_s = 0.015$ source error rate, $\beta = 3.2 \times 10^{-7}$ /km fibre dephasing, $\eta_{bs} = 0.97$ beam-splitter efficiency, Charlie at the arm midpoint), with 1000 Monte Carlo runs per point and shaded bands showing the sample spread.

Three groupings emerge from these seven sweeps, extending the channel-versus-per-arm pattern established in Section 4.1.

Channel-side loss. Fibre attenuation (Figure 14a) behaves exactly like the distance sweep of Section 4.1: the BB84/MDI ratio stays close to $5.2\text{--}5.3\times$ across the full 0–0.3 dB/km range, confirming that fibre attenuation, like distance, acts on the shared channel before Alice’s and Bob’s photons diverge towards Charlie, so it scales both protocols together rather than widening the gap between them.

Per-arm hardware loss. Beam-splitter efficiency (Figure 14b) and TX insertion loss (Figure 14c) both act after the split, and both widen the ratio, consistent with detector efficiency and receiver loss in Section 4.1. Beam-splitter efficiency is unique among all eleven sweeps in that BB84 is structurally unaffected by it (the dashed reference line in Figure 14b is exactly flat, since η_{bs} only enters MDI’s Bell-state measurement); MDI key rate alone declines from 66.1 kbps at $\eta_{bs} = 1.0$ to 42.3 kbps at $\eta_{bs} = 0.8$, a 36% reduction from a single power of η_{bs} in the coincidence probability. TX insertion loss widens the BB84/MDI ratio from $4.8\times$ at $L_i = 0$ to $9.7\times$ at $L_i = 0.5$, the same per-arm doubling pattern seen under detector efficiency (Figure 6) and receiver loss (Section 4.1). X-basis detector efficiency, or basis bias (Figure 14d), belongs to this group too but widens the ratio only modestly, from $5.0\times$ at $\eta_X = 1.0$ to $6.1\times$ at $\eta_X = 0.5$, far less than the full detector-efficiency sweep of Figure 6, because here only one of the two measurement bases is degraded rather than both.

Noise-like parameters. Dark count rate, fibre dephasing rate, and source error rate leave both protocols essentially flat across their full swept ranges at this metropolitan oper-

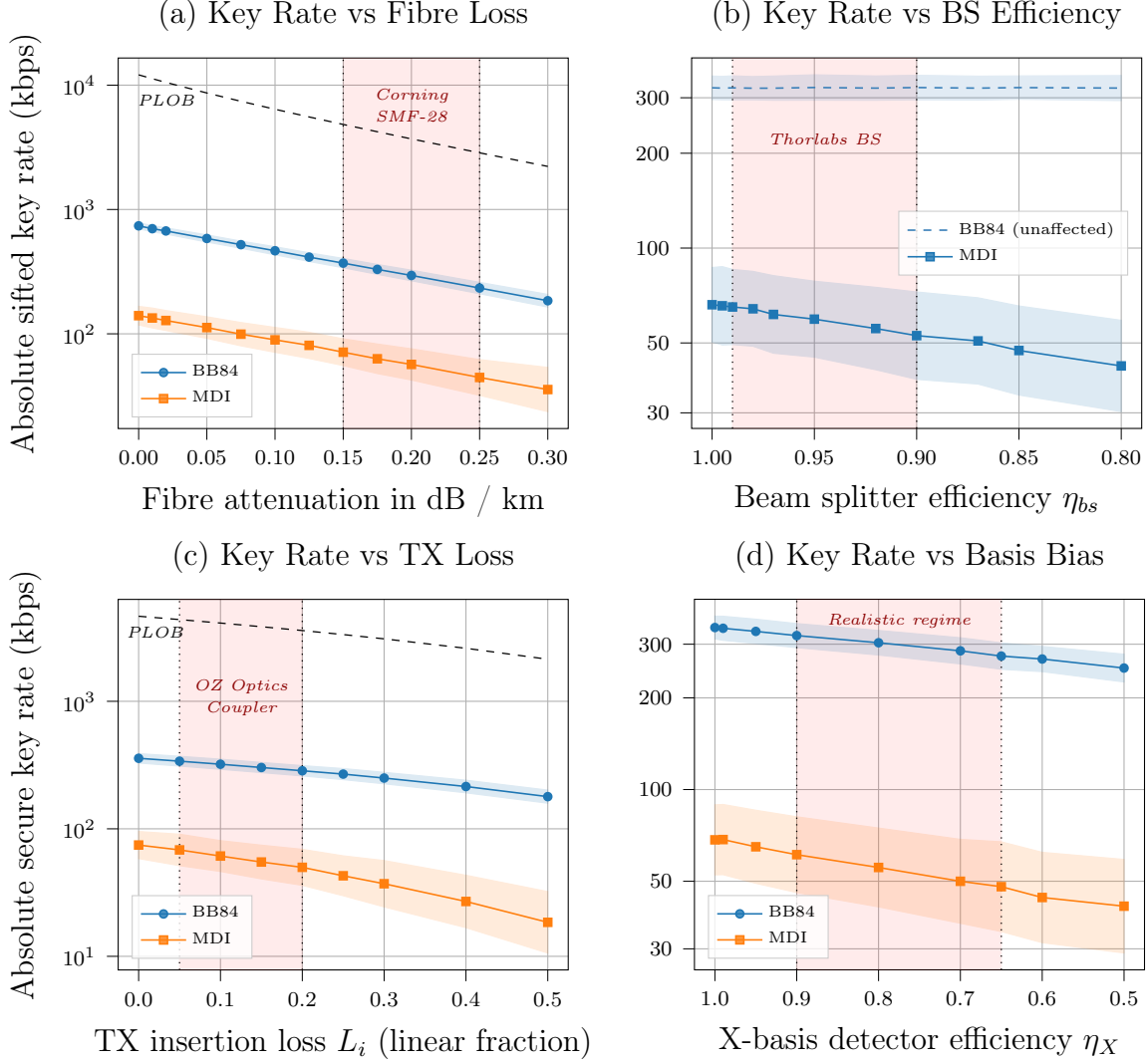
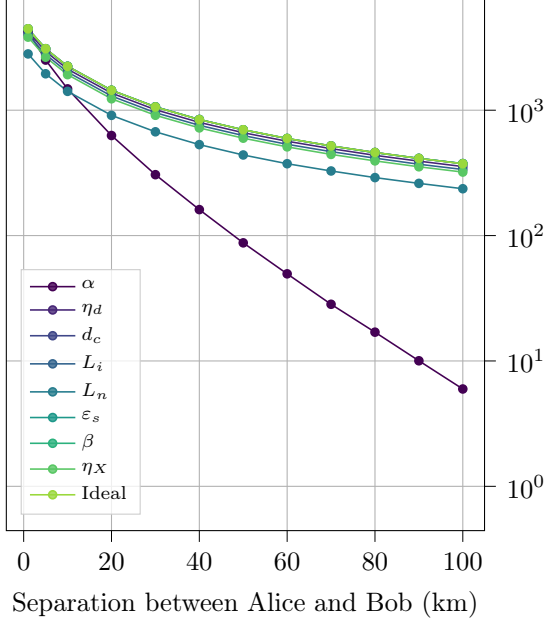


Figure 14: Four of the remaining point-to-point parameter sweeps (see *Noise-like parameters* below for the other three). BB84 in blue (circles), MDI in orange (squares); shaded bands show Monte Carlo sample spread. Except for the parameter swept in each panel, all hardware parameters take their layer-9 default values from Table 2.

ating point, with the BB84/MDI ratio staying within a few percent of $5.2\times$ throughout each; flat enough that the three are omitted from Figure 14 and reported numerically here instead. Across each parameter’s realistic hardware range, key rate varies from best to worst point by: dark count rate, 0.8% (BB84) and 1.9% (MDI); fibre dephasing rate, 0.6% (BB84) and 2.0% (MDI); source error rate, 0.7% (BB84) and 3.0% (MDI). None of the three introduces a directional loss term on either arm, so none is expected to shift the ratio; as noted in Section 5.3, this flatness is specific to the signal rates achieved at 20 km and should not be extrapolated to longer, detection-starved links.

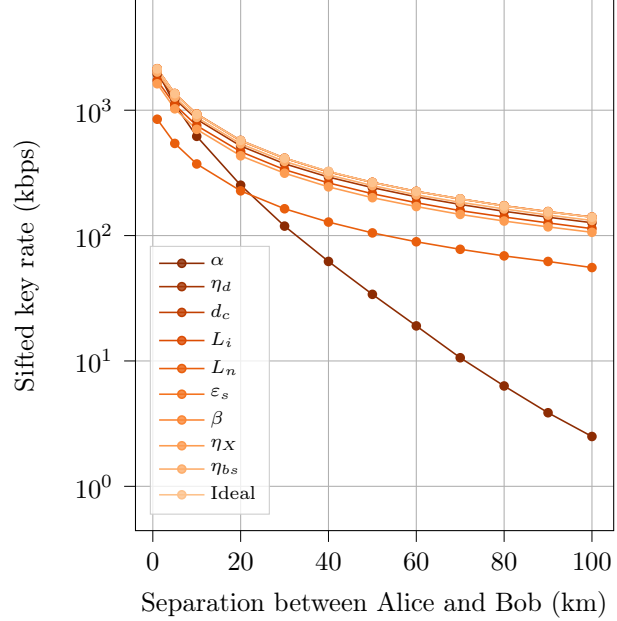
Isolated parameter impact. Figure 15 extends the fixed-distance ranking of Figure 10 (Section 4.1) across the full 1–100 km range, setting exactly one parameter to its value at a time against an otherwise fully ideal baseline.

BB84: Isolated Parameter Impact



(a) BB84

MDI: Isolated Parameter Impact



(b) MDI

Figure 15: Sifted key rate versus Alice–Bob separation with exactly one hardware parameter set to its fully realistic value (Table 2), against an ideal (baseline) reference. Curves show the Monte Carlo mean.

The two parameters singled out in Section 4.1 behave structurally differently across distance. Fibre loss tracks the ideal baseline closely at short range, pulling away with distance (BB84 reduction widens from 4.3% at 1 km to 98.4% at 100 km, MDI from 3.7% to 98.2%). Node loss instead reduces key rate by a *near-constant fraction at every distance*: 36.9% at 1 km versus 36.8% at 100 km for BB84, and 60.3% at both distances for MDI, matching the 25 km sensitivity figures to within Monte Carlo noise. Node loss, basis bias, TX insertion loss, and (for MDI) beam-splitter efficiency therefore constitute a qualitatively distinct category of impairment from fibre attenuation: fixed-fraction, per-arm penalties eroding key rate independent of separation and consistently favouring BB84, rather than range-dependent losses, equivalent between protocols, only becoming significant as a link extends.

B Relay Placement Optimisation

Section 3.1.4 proposed replacing the plain k -means-centroid relay placement used elsewhere in this work’s development with a generalised Weiszfeld iteration, on the grounds that it correctly minimises total fibre length, rather than the sum of squared distances that k -means centroids minimise. This appendix works through that argument in more detail: it derives the coupled update of Equation 7 from first principles, works a small example to build intuition for how and why the two objectives diverge, and reports a self-contained geometric comparison of the two placement methods across synthetic topologies matching those used in Section 3.1.4. This comparison is purely geometric – total fibre length under each placement; the discussion questions at the end of this appendix are correspondingly open rather than resolved.

Derivation of the coupled update. This is the classical multi-facility Weber problem, dating to Miehe’s original network-of-junctions formulation [61] and specialised to hub-and-spoke interconnection by O’Kelly [62]; Alumur and Kara [63] survey the broader hub-location literature that has since grown around it. Recall the total fibre length objective from Equation 6,

$$L(x_1, \dots, x_K) = \sum_{k=1}^K \sum_{i \in C_k} \|x_k - p_i\| + \sum_{k < l} \|x_k - x_l\|,$$

where C_k is the fixed set of users assigned to relay k and x_k its free position. Away from the (measure-zero) configurations where x_k coincides with a user p_i or another relay x_l , L is differentiable, and each term contributes a unit vector pointing from the fixed point towards x_k , scaled by the reciprocal distance:

$$\nabla_{x_k} L = \sum_{i \in C_k} \frac{x_k - p_i}{\|x_k - p_i\|} + \sum_{l \neq k} \frac{x_k - x_l}{\|x_k - x_l\|}.$$

Setting $\nabla_{x_k} L = 0$ and separating the x_k -weighted terms from the fixed-point-weighted terms on each side gives

$$x_k \left(\sum_{i \in C_k} \frac{1}{\|x_k - p_i\|} + \sum_{l \neq k} \frac{1}{\|x_k - x_l\|} \right) = \sum_{i \in C_k} \frac{p_i}{\|x_k - p_i\|} + \sum_{l \neq k} \frac{x_l}{\|x_k - x_l\|},$$

which rearranges directly to the fixed-point update of Equation 7 by evaluating the right-hand side at iteration t and solving for x_k at iteration $t+1$. This is a majorize-minimise step, not a gradient step: each relay is repositioned to the weighted mean of the points pulling on it, with weights set by the *previous* iterate’s distances, and iterating is guaranteed to be

non-increasing in L (Kuhn [48]) even though no step size is chosen explicitly.

The degenerate case, where an iterate $x_k^{(t)}$ lands exactly on a p_i or x_l , produces a division by zero, since the corresponding term is non-differentiable there (the norm has a subgradient of unit magnitude in any direction rather than a unique gradient). In practice, and in the numerical experiments below, this is handled by flooring every pairwise distance at a small ϵ (we use $\epsilon = 10^{-9}$ km) before taking its reciprocal; this perturbs the objective negligibly at realistic network scales but avoids the singularity entirely. A more principled fix – checking the subgradient optimality condition explicitly at any iterate landing within ϵ of a fixed point, as in the standard single-facility Weiszfeld literature – was not needed at the N, K scales explored here, since exact coincidence between a continuous-uniform random draw and another point has probability zero and was never observed in $30 \times 9 = 270$ seeded topology trials.

Worked example: centroid vs. geometric median. A small hand-constructed instance builds intuition for the divergence between the two objectives: eight users in two well-separated groups of four, one of which contains a single point noticeably further from the other three than they are from each other. The k -means centroid of that cluster is pulled towards this one distant point, since it enters the sum-of-squares objective quadratically; the geometric median instead settles closer to the tight group of three, since the same point enters the sum-of-*unsquared*-distances objective only linearly. This is the standard robustness property of the median relative to the mean, familiar from one-dimensional statistics, applied here to a two-dimensional facility-location setting. The effect on total fibre length is modest but directionally consistent with every trial in the larger comparison below: 15.53 km under the k -means centroid versus 14.62 km under the coupled Weiszfeld update, a 5.9% reduction, obtained without changing which users are assigned to which relay (cluster assignment is identical under both placements).

Geometric comparison across network topologies. To quantify the fibre-length reduction at realistic network scales, both placement methods were run on synthetic topologies generated identically to those used in Section 3.1.4: N users placed uniformly at random within a 10×10 km service area, for $N \in \{10, 20, 50\}$ and relay counts $K \in \{2, 3, 5\}$, over 30 independent seeds per (N, K) pair. For each seed, k -means clustering ($n_{\text{init}} = 10$) provides both the baseline centroid placement and the initial cluster assignment for the coupled Weiszfeld refinement (Equation 7), which then alternates position updates with nearest-relay reassignment until total fibre length changes by less than 10^{-9} km between iterations (typically several dozen iterations; up to 146 in the smallest, most relay-dense case). Table 5 reports total fibre length under each method, averaged across seeds.

N	K	k -means length (km)	Weiszfeld length (km)	Reduction
10	2	29.95	28.57	4.6% $\pm$ 2.7
10	3	35.42	30.92	12.6% $\pm$ 3.8
10	5	69.58	37.89	45.6% $\pm$ 3.7
20	2	57.23	56.00	2.1% $\pm$ 1.4
20	3	56.31	53.61	4.8% $\pm$ 1.8
20	5	84.80	67.13	20.8% $\pm$ 4.6
50	2	143.01	142.11	0.6% $\pm$ 0.4
50	3	124.03	122.06	1.6% $\pm$ 0.8
50	5	133.63	126.01	5.7% $\pm$ 1.9

Table 5: Mean total fibre length (spoke + backbone) under k -means centroid versus generalised Weiszfeld relay placement, across 30 seeds per (N, K) pair, 10×10 km service area. Reduction is $100(L_{\text{kmeans}} - L_{\text{Weiszfeld}})/L_{\text{kmeans}}$, mean $\pm$ one standard deviation across seeds.

The reduction is positive in every trial (consistent with the monotonic non-increase guarantee noted above, since the Weiszfeld iteration starts from the k -means solution and never leaves it worse off) and grows sharply with K at fixed N : from 4.6% at $K = 2$ to 45.6% at $K = 5$ for $N = 10$, and from 0.6% to 5.7% across the same K range for $N = 50$. This follows directly from Equation 6: the backbone term has $K(K - 1)/2$ pairwise contributions, growing quadratically in K , while the spoke term grows only linearly in N ; at larger K the backbone increasingly dominates total cost, and it is precisely the backbone coupling that a per-cluster centroid ignores entirely. Conversely, the reduction shrinks as N grows at fixed K : from 45.6% to 5.7% at $K = 5$ between $N = 10$ and $N = 50$. A larger user base dilutes each relay’s positioning error relative to the now spoke-dominated total, and tighter, more populated clusters leave less room for a single geometric outlier to pull the centroid away from the coupled optimum. The practical implication is that this refinement matters most exactly where the MDI network topology already carries a real relay count K relative to N – the regime the cost-crossover analysis of Section 4.3 identifies as commercially relevant for smaller deployments before BB84’s $\mathcal{O}(N^2)$ fibre term dominates – and least where N is large enough that per-user fibre cost already dominates the network total regardless of relay placement.

Figure 16 shows this mechanism directly at $N = 20$, $K = 3$ (3.7% reduction, close to this cell’s 4.8% seed-averaged mean in Table 5): the Weiszfeld backbone triangle (right) is visibly tighter than the k -means triangle (left), each relay pulled a short distance towards the other two, while spoke assignment and per-cluster relay position barely move.

Open questions. The comparison above resolves the geometric question: which placement

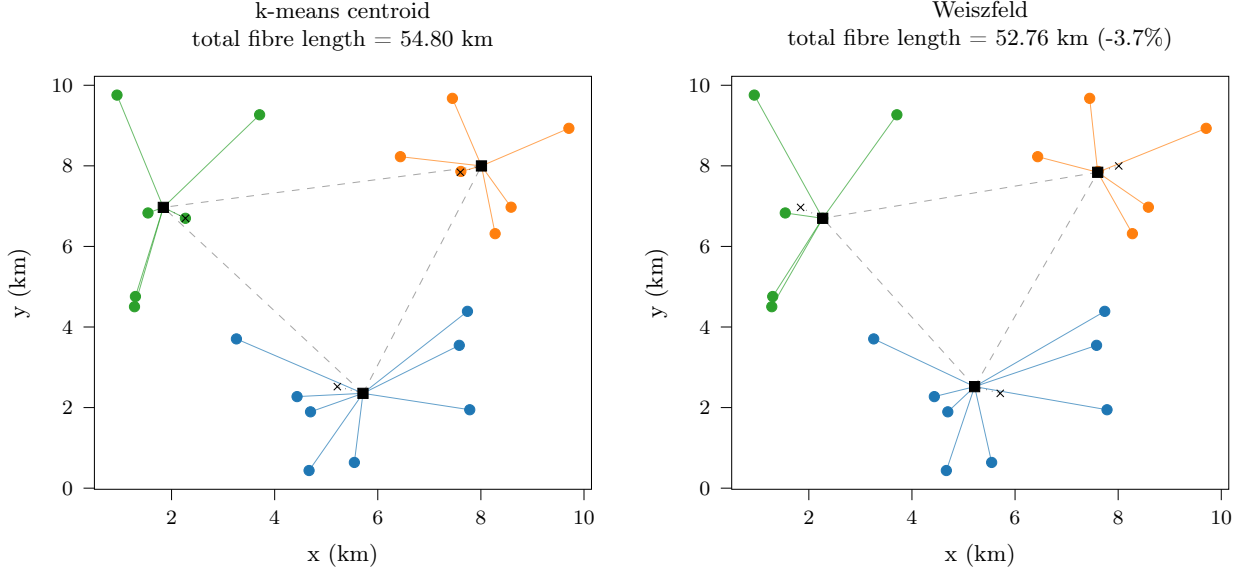


Figure 16: Relay placement on a shared $N = 20$ -user layout, $K = 3$ relays: k -means centroid (left) versus generalised Weiszfeld (right). Circles are users, coloured by cluster; squares mark each panel’s own relay positions, joined by the dashed backbone; crosses mark the *other* panel’s relays for comparison.

minimises total fibre length for a *fixed* backbone topology and cluster assignment? However, it still leaves several modelling questions open, appropriate for discussion before the new placement method is adopted for the network-level key rate sweeps of Section 3.1.4:

- *Backbone topology.* Table 3 assumes a full $K(K - 1)/2$ relay mesh. A minimum spanning tree backbone would in general have strictly lower total length for $K > 2$, and would change the placement problem itself (the coupling term in Equation 6 would run over tree edges rather than all pairs). Whether the full mesh is retained for redundancy/latency reasons independent of cost should be decided before the cost model and placement method are finalised together.
- *Assignment optimality.* Cluster reassignment after each position update is by nearest Euclidean distance to a relay, which is the correct assignment rule for the spoke term alone but does not account for the backbone term: a user could in principle be assigned to a farther relay if doing so allowed both relays to move to a jointly cheaper configuration. This is a combinatorial partitioning question distinct from the continuous position optimisation solved by Equation 7, and was not explored here.
- *Convergence cost.* The Weiszfeld refinement took up to 146 outer iterations to reach a 10^{-9} km convergence tolerance at the smallest, most relay-dense configuration tested ($N = 10$, $K = 5$); this is negligible against NetSquid’s own per-trial cost (Appendix D) but would be worth profiling at the $N = 100$ scale flagged there as currently infeasible.

- *Net effect on key rate.* A shorter mean path length should translate into a modest key rate improvement through reduced fibre attenuation (Section 3.1.2, layer 1), but by how much depends on where in the loss-versus-distance curve of Figure 9 the affected links sit; this can only be answered by re-running the network-level sweeps of Section 3.1.4 under the new placement, which had not yet been done at the time of writing.
- *Degenerate coincidence.* Equation 7 (and its $K = 1$ case) divides by the distance between the current iterate and each fixed point, which is undefined if a relay iterate ever coincides exactly with a user (or, in the multi-facility case, another relay). The implementation guards against this by flooring the distance at 10^{-9} km rather than applying Ostresh’s modification – which tests whether the coincident point is itself already a stationary point of the weighted sum and, if not, steps away from it along a descent direction. The floor avoids a division-by-zero crash but is not equivalent: it lets that term’s weight grow to $\mathcal{O}(10^9)$, which can pin the next iterate back near the same point rather than resolving the singularity correctly. Coincidence is measure-zero under the continuous uniform and Gaussian user placements used throughout this work, so it was not observed to trigger in practice, but the implementation should not be trusted as-is on discretised or clustered location data (e.g. users snapped to a road network, or a relay initialised exactly at a fixed real-topology co-ordinate) where exact coincidence is plausible.

C Real-World Deployment Considerations

Every network-scale result in the main text uses synthetic user and relay placements, uniformly or Gaussian-distributed over an idealised square service area. A real provider does not have this freedom: relay sites are often constrained to existing telecommunications infrastructure rather than an analytically convenient location, and fibre routes follow duct and roadway paths rather than straight-line geometry. This appendix discusses how the protocol and cost-modelling choices made throughout this work adapt to that constraint, illustrated with a small case study on a real topology: three sites from BT’s London Quantum-Secured Metro Network [47] (Slough, West End, City of London) taken as $K = 3$ relay locations, with synthetic users anchored around them, since no real user-site data exists.

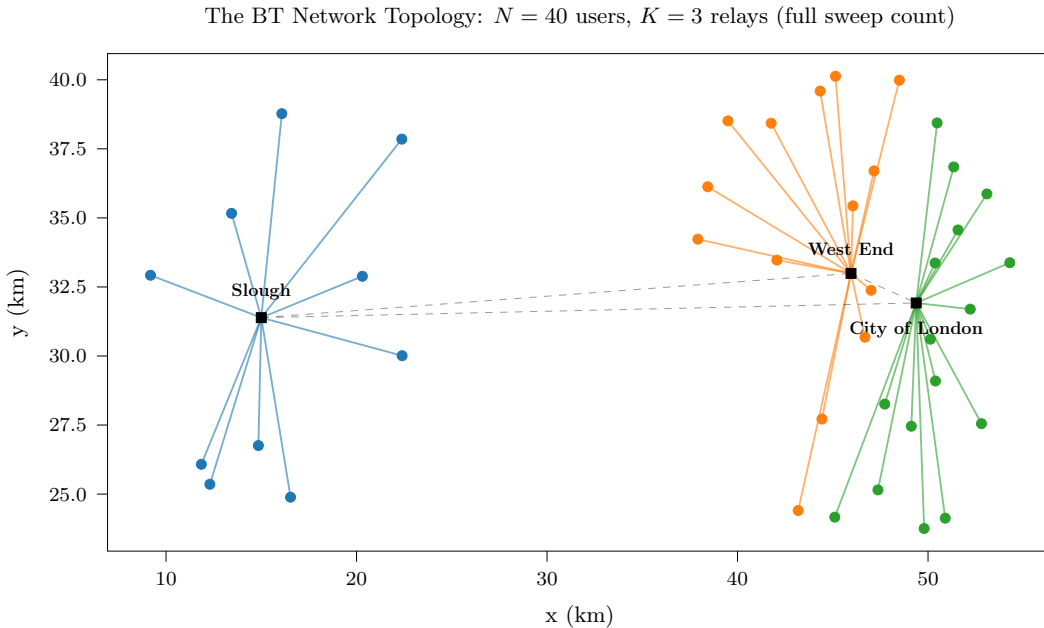


Figure 17: The real BT site layout used for the case study below, at the largest user count tested. Relay markers map the three real site coordinates; users are synthetic, anchored on the three real sites; dashed lines are the $K(K - 1)/2$ backbone mesh.

Protocol choice under a fixed, real topology. The three network protocols implemented in this work (Section 3.1.4) trade range for trust in different ways once relay and user locations are fixed by real geography rather than drawn synthetically. Direct BB84 needs no relay infrastructure at all, but its reach is capped by loss budget alone, so a provider is limited to serving user pairs that happen to fall within range of each other. MDI-QKD extends coverage through untrusted Bell-state relays sited wherever suitable premises exist, at the key-rate cost quantified throughout this work. TBB84 extends coverage further still by chaining relay hops, at the cost of trusting every intermediate site with plaintext-equivalent key material; this is

the operational model already used by existing metropolitan QKD testbeds [8,9,46], making it the most field-proven, if least novel, of the three.

The BT case study: key rate. Unlike the cost figures below, a key rate for this topology requires actually running NetSquid rather than evaluating a closed form, so this figure is confined to a narrower $N = 5\text{--}15$ sweep (organic growth on the fixed $K = 3$ real placement, catchment radius 10 km, seed 42) to keep wall-clock cost tractable (Appendix D), at $r = 1000$ trials per pair – the same budget already justified for the bipartite sweeps of Section 4.1, whose worst-case CV gives $\text{SEM} \approx 1\%$, comfortably below the effect sizes reported here. With no seed-averaging to fall back on (this is one fixed real topology, not a distribution over drawn ones), trial-level sampling noise is the only source of uncertainty in Figure 18’s curves.

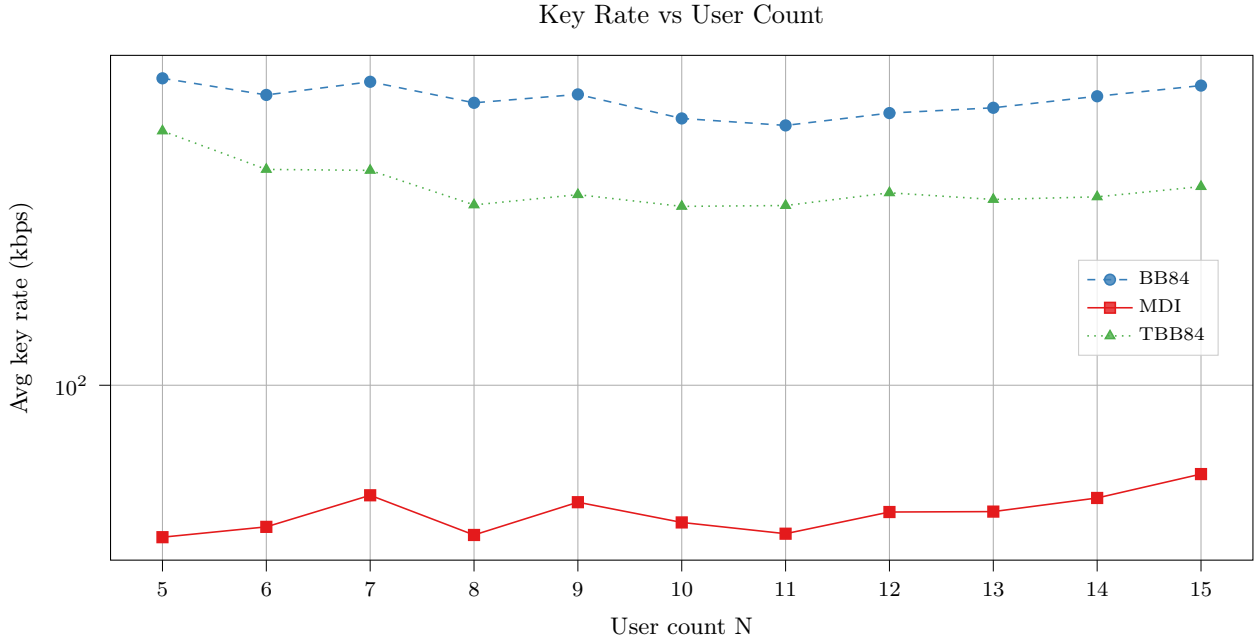


Figure 18: Average key rate against user count on the real BT topology, $K = 3$, $r = 1000$ trials per pair. Single fixed topology, not seed-averaged.

The ranking matches the synthetic-topology results throughout Section 4: BB84 highest (426–554 kbps), TBB84 second (271–413 kbps), MDI lowest (43–61 kbps) at every N tested. The BB84/MDI gap is narrower here than the bipartite $\sim 5\text{--}25\times$ range of Section 5.1 would suggest on its own – roughly $9\text{--}13\times$ across the sweep – consistent with this topology’s short West End–City of London leg (3.4 km) pulling the relay-routed protocols’ effective per-pair distance down for a large share of user pairs. TBB84 sits between the other two throughout, as expected: it inherits BB84’s single-photon detection at each hop rather than MDI’s two-photon coincidence penalty, but pays a multi-hop tax BB84’s direct mesh does not, so it trades some of BB84’s rate for MDI’s reach without needing MDI’s relay trust model to get

there.

Cost modelling under real prices. The symbolic cost model used throughout this work (Section 4.3) leaves component prices as free parameters, keeping its conclusions valid across vendors and over time, but it cannot produce an absolute cost figure for a specific deployment. A fully numeric model with vendor-quoted prices could produce that figure, at the cost of being tied to prices that are commercially sensitive, quickly outdated, and, as encountered during this work, difficult to pin down with enough confidence to publish as validated rather than illustrative. The case study below takes the illustrative middle path: fixed indicative prices, real fibre lengths.

The BT case study: cost. Using the real fibre lengths between the topology of Figure 17’s sites and their assigned users – not the closed-form geometric-probability approximation of Equations 10–12 – alongside illustrative component prices ($c_s = \text{£}250\text{k}$ per source, $c_d = \text{£}100\text{k}$ per detector module, $c_f = \text{£}10\text{k}/\text{km}$ new-build dark fibre) gives Figure 19. TBB84’s cost is defined identically to MDI’s throughout this work (same component counts under the multi-channel-detector simplification of Section 4.3, differing only in trust assumptions, not hardware), so the two curves coincide exactly by construction, not coincidence.

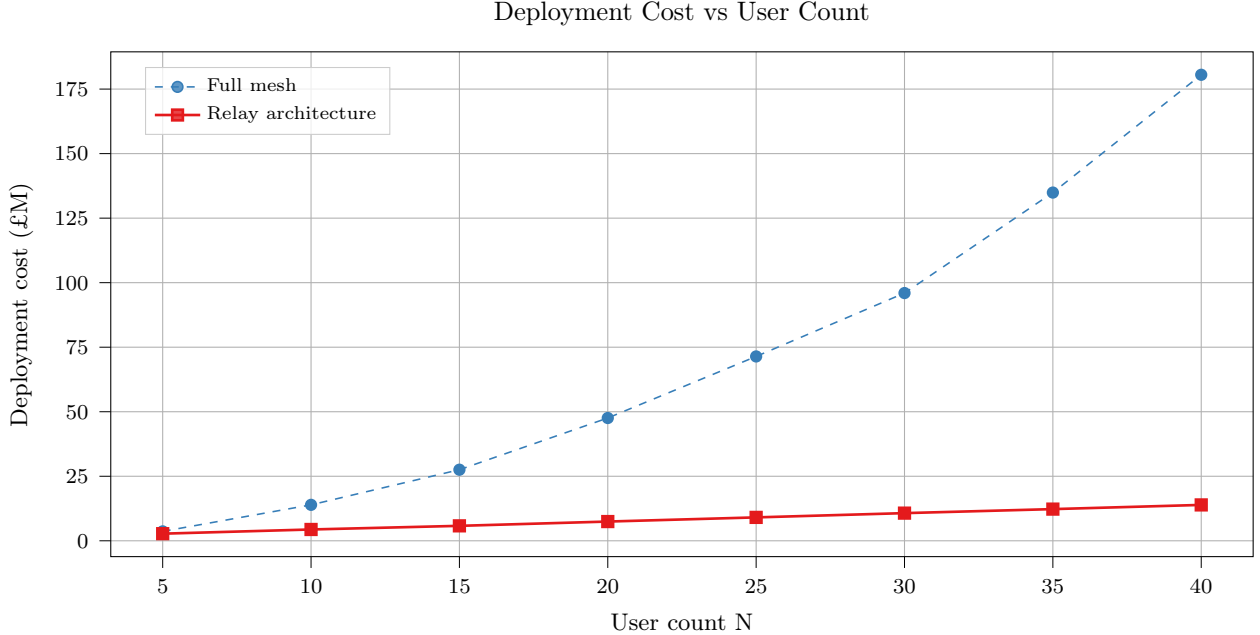


Figure 19: Deployment cost against user count on the real BT topology, $K = 3$, illustrative prices (text). “Full mesh” is BB84’s direct fibre mesh; “relay architecture” is MDI and TBB84, which coincide exactly (see text).

BB84’s direct mesh is already the most expensive option at the smallest N tested and diverges sharply beyond it (£3.6M to £180.5M, $N = 5$ to 40), against MDI/TBB84’s near-

linear £2.8M to £13.9M – a twelvefold gap by $N = 40$, on real, non-uniform geography rather than the idealised uniform scattering Section 4.3’s closed form assumes.

The BT case study: cost-efficiency. Combining Figure 18’s rates with Figure 19’s costs at the shared $N = 5$ –15 range gives Figure 20, key rate per unit cost, the single number a customer sizing a deployment against a fixed budget would actually want.

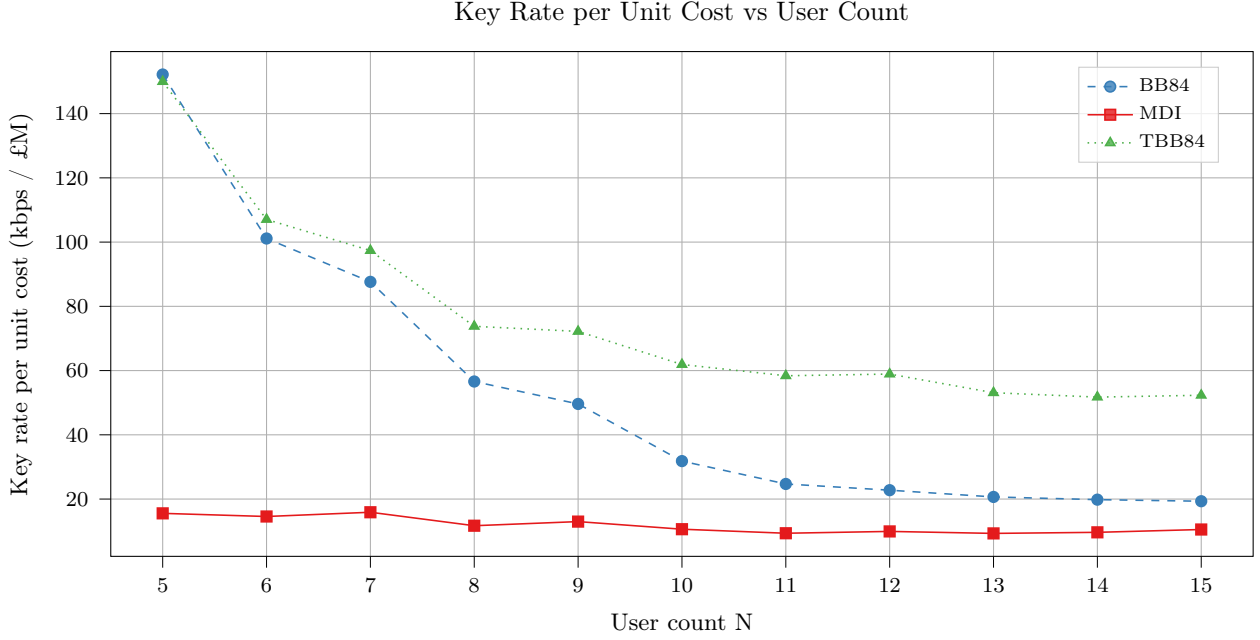


Figure 20: Key rate per unit cost against user count on the real BT topology, $K = 3$. Derived from Figures 18 and 19 at their shared N range. Single fixed topology, not seed-averaged.

For $N \geq 6$, cost-efficiency ranks the three protocols in exactly the reverse order of the trust hierarchy of Section 4.4: TBB84 (Level 3, trust everything) is most cost-efficient, BB84 (Level 2) sits in the middle, and MDI (Level 1, least trusting) is least cost-efficient throughout, by roughly an order of magnitude against either alternative. The one exception is $N = 5$, where BB84 (152.2 kbps/£M) edges out TBB84 (150.0 kbps/£M) by under 1.5% – close enough to read as a tie rather than a genuine crossover, given the trial-level sampling noise discussed above. The headline is therefore not simply “cheaper is better”: it is that this real topology reproduces, almost exactly, the inverse relationship between trust and cost-efficiency the security discussion of Section 5.1 already predicts on different grounds – MDI’s device-independence is bought with both a rate penalty and a cost-efficiency penalty, not a rate penalty alone.

Figure 21 extends the cost comparison beyond this narrower rate-limited range, as a ratio, $C_{\text{relay}}/C_{\text{BB84}}$, swept across eight values of $\rho = c_d/c_F$ spanning and extending Figure 13’s range, with $K = 3$ held fixed throughout to match BT’s actual relay count.

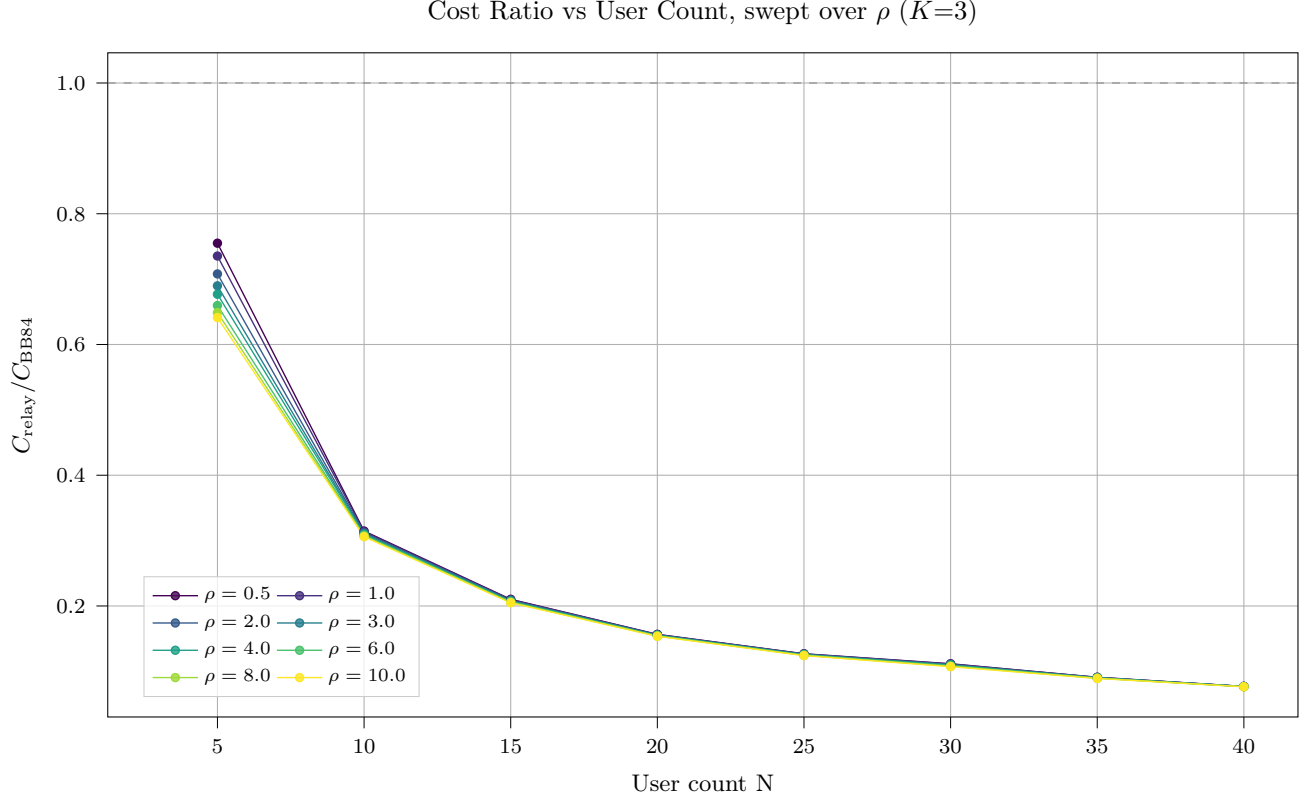


Figure 21: Cost ratio $C_{\text{relay}}/C_{\text{BB84}}$ against user count on the real BT topology, $K = 3$, across eight ρ values. Curves are deterministic given N , K , ρ , and the fixed real distances (no Monte Carlo noise). Single fixed topology, not seed-averaged.

Every curve is already below 0.75 at $N = 5$, the smallest user count tested – consistent with N^* 's closed-form confinement to $(K, K+1)$ for any ρ (Section 4.3), the crossover on this real topology falls before this figure's practically-relevant sweep even begins, well below any N a provider would plausibly size $K = 3$ relay infrastructure for. More informative is how quickly ρ stops mattering: the eight curves are visually indistinguishable beyond $N \approx 15$, converging to within a few percent of one another by $N = 40$. Past that point the fibre term dominates total cost regardless of detector price, the same mechanism behind $N^* \rightarrow K$ as $\rho \rightarrow \infty$ in Equation 16, now visible directly in the cost ratio itself rather than only in the crossover point, and on a real rather than idealised topology.

A rigorous treatment of vendor-price-anchored cost estimation, and of relay-site-constrained protocol selection more generally, is left as future work (Section 6.3).

D NetSquid: Limitations and Speedup Strategies

Section 5.3 flags NetSquid’s computational cost as a limitation that constrained the seed and Monte Carlo budgets feasible within the project timeline. This appendix details that cost, the mitigation strategies attempted during this work, and the further engineering a commercial-scale successor would plausibly need.

The i.i.d. modelling assumption. The simulations in this work model each photon trial as an independent, identically distributed (i.i.d.) draw: no channel memory, no inter-trial correlations, no time-varying network state. Under this assumption, the key rate distributions produced by NetSquid are in principle reproducible by a simple analytical probability model operating on the same physical parameters, and NetSquid’s overhead – launching a DES engine, propagating quantum state objects, managing event queues – is not strictly necessary for the i.i.d. regime. The justification for using NetSquid regardless is extensibility: the DES framework supports temporally correlated dynamics (detector deadtime, jitter, burst traffic, entanglement distillation protocols) that an i.i.d. model cannot accommodate, and these effects become important if this infrastructure is extended to quantum repeater chains or adaptive network protocols (Section 6.3).

Wall-clock cost. The computational cost of the DES approach is non-negligible: network sweeps across multiple seeds and user counts required on the order of hours of wall-clock time on a multi-core workstation. This constrained the number of seeds and Monte Carlo runs feasible within the project timeline, and limits the statistical confidence of the topology-averaged results. Table 6 summarises representative wall-clock costs on a multi-core workstation: a full point-to-point parameter sweep completes in minutes, but cost grows sharply with network size, reaching hours for a $N=20$ relay sweep and an estimated weeks for $N=100$ – infeasible within the project timeline. The dominant cost drivers are the $\mathcal{O}(N^2)$ user-pair count, the sequential causal event loop within each NetSquid run (which cannot be parallelised across a single trial), and multiprocessing spawn overhead of approximately 2–3 s per Monte Carlo batch, independent of workload.

Scenario	Wall-clock time	Feasibility
P2P sweep	~minutes	OK
$N=20$ relay sweep	~hours	Marginal
$N=100$ sweep	~weeks	Infeasible

Table 6: Wall-clock time for representative NetSquid sweep scenarios on a multi-core workstation.

Mitigation strategies attempted. Two mitigations were investigated during this work. Adaptive Monte Carlo sampling terminates trials early once the running key-rate estimate has converged, reserving the full sample budget for slower-converging points; this reduced BB84 wall-clock time by 30% (28% fewer runs, <1% relative error) but gave negligible savings for MDI, whose structurally higher variance (Section 4.1, *Sample spread*) rarely triggers early stopping. Gaussian process surrogate modelling instead trains a regression model on a coarse NetSquid-generated grid and evaluates a dense 10^4 -point grid in under 100 ms, giving near-instant interpolation once trained; however, it requires a substantial upfront training sweep, its accuracy degrades outside the training range, and it defers rather than reduces the underlying simulation cost. Neither approach addresses the root cause, since cost is set by control flow and process management (the sequential event loop and spawn overhead) rather than floating-point throughput – for the same reason, GPU acceleration was considered and rejected, as BB84/MDI operate on 1–2-qubit density matrices (2×2 or 4×4) too small for GPU parallelism to outweigh its overhead, and NetSquid has no GPU backend regardless. These constraints motivated the relay-count and user-count ranges explored in Section 3.1.4.

Proposed speedup strategies. The bottlenecks identified above are engineering problems rather than fundamental limits of the discrete-event approach, and none of the following was attempted here for want of time. Batched or vectorised trial execution would amortise spawn cost and interpreter startup across many trials rather than paying it once per Monte Carlo batch. A persistent worker pool that keeps interpreter and NetSquid state warm across sweep points would remove the 2–3 s per-batch spawn overhead entirely, in place of the spawn-per-batch model used throughout this work. For the sequential per-trial event loop itself, a shared-memory or distributed DES backend could parallelise causally independent sub-chains of a single network-scale trial (e.g. non-interacting user pairs), rather than only parallelising across whole trials as done here. Classical network simulation faced an analogous scaling problem before tools such as ns-3 [27] and OMNeT++ [28] matured into viable commercial-scale planning tools; the wall-clock costs quantified in Table 6 are offered as a first characterisation of where NetSquid’s current execution model breaks down, for a comparable QKD-specific tool to build on. Read this way, the open question raised by this appendix is not whether NetSquid’s general-purpose discrete-event engine is the right abstraction for network-scale QKD simulation in principle, but what engineering investment is required to make it practical at the network sizes a commercial deployment-planning tool would need to support.

E Code and Reproducibility

Availability. All simulation, network, cost-model, and plotting code is publicly available at <https://github.com/linuskelsey/netsquid-qkd>, released under a permissive MIT-style licence. The repository is organised into `BB84/` and `MDI/` (protocol implementations and simulation runners), `network/` (the multi-user topology layer and BB84/MDI/trusted-BB84 network simulators), `lib/` (shared configuration loading, database, and plotting utilities), `configs/` (the ten JSON hardware-layer presets of Table 2), `scripts/P2P/` and `scripts/network/` (the point-to-point and network-scale sweep drivers behind every figure in this report), `scripts/analyse/` (figure reconstruction from stored results, described below), and `docs/` (this dissertation’s $\text{\LaTeX}$ source and figures). Every script exposes a `-help` option and carries a usage docstring; running any figure-producing script end to end requires only `pip install -r requirements.txt` plus a free academic registration with NetSquid itself at netsquid.org, since NetSquid is not distributed on the public PyPI index.

Results database. Every Monte Carlo batch – whether from a point-to-point sweep, a network-scale sweep, or an ad-hoc script invocation – is persisted to a local SQLite database (`data/results.db`) via `lib/db.py`, rather than being discarded once a figure is drawn. The full schema is documented in `DATA.md`. This design decouples analysis from re-simulation: `scripts/analyse/` can reconstruct any figure in this report directly from the database, re-running only the aggregation and plotting code, which is what made it practical to revisit plot formatting, error-display mode, and shaded-region annotations (Section 4.1) without re-incurring the wall-clock costs of Appendix D. The database itself is not version-controlled (`data/` is git-ignored): it is regenerated by construction on first run and grows unboundedly as further sweeps are executed, so committing it would mean committing an ever-growing, non-reviewable binary artefact rather than the code that produces it. The raw database backing the results in this report is available on request.

Figure provenance. Every figure in this report also has its full provenance preserved under `docs/figures/August/final-figs/` in the repository. Each figure’s directory contains an `assumptions.md` file recording every fixed parameter used to produce it (seed, user and relay counts, service area, tortuosity, runtimes per pair, and the resulting key rates), alongside the rendered plot and its $\text{\LaTeX}$ source. Network-scale figures additionally include a `topologies/` subdirectory with the exact relay and user placement underlying each data point (one image per relay count and protocol) so the specific geometry behind any reported key rate can be inspected directly rather than only its aggregate statistics.

References

- [1] R. Holz *et al.*, “The SSL landscape: a thorough analysis of the x.509 PKI using active and passive measurements,” in *Proceedings of the 2011 ACM SIGCOMM Conference on Internet Measurement Conference*, pp. 427–444, 2011.
- [2] R. L. Rivest, A. Shamir, and L. Adleman, “A method for obtaining digital signatures and public-key cryptosystems,” *Commun. ACM*, vol. 21, p. 120–126, Feb. 1978.
- [3] N. Koblitz, “Elliptic Curve Cryptosystems,” *Mathematics of Computation*, vol. 48, pp. 203–209, Jan. 1987.
- [4] V. S. Miller, “Use of Elliptic Curves in Cryptography,” in *Advances in Cryptology — CRYPTO ’85 Proceedings* (H. C. Williams, ed.), (Berlin, Heidelberg), pp. 417–426, Springer Berlin Heidelberg, 1986.
- [5] P. W. Shor, “Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer,” *SIAM Journal on Computing*, vol. 26, p. 1484–1509, Oct. 1997.
- [6] S. Wiesner, “Conjugate coding,” *SIGACT News*, vol. 15, p. 78–88, Jan. 1983.
- [7] C. H. Bennett and G. Brassard, “Quantum cryptography: Public key distribution and coin tossing,” *Theoretical Computer Science*, vol. 560, p. 7–11, Dec. 2014.
- [8] Y.-A. Chen *et al.*, “An integrated space-to-ground quantum communication network over 4,600 kilometres,” *Nature*, vol. 589, 01 2021.
- [9] M. Peev *et al.*, “The SECOQC quantum key distribution network in Vienna,” *New Journal of Physics*, vol. 11, p. 075001, 07 2009.
- [10] M. Sasaki *et al.*, “Field test of quantum key distribution in the Tokyo QKD Network,” *Optics Express*, vol. 19, p. 10387, May 2011.
- [11] H.-K. Lo, M. Curty, and B. Qi, “Measurement-Device-Independent Quantum Key Distribution,” *Phys. Rev. Lett.*, vol. 108, p. 130503, Mar 2012.
- [12] L. Lydersen, C. Wiechers, C. Wittmann, D. Elser, J. Skaar, and V. Makarov, “Hacking commercial quantum cryptography systems by tailored bright illumination,” *Nature Photonics*, vol. 4, no. 10, pp. 686–689, 2010.

- [13] V. Makarov, “Controlling passively quenched single photon detectors by bright light,” *New Journal of Physics*, vol. 11, no. 6, p. 065003, 2009.
- [14] I. Gerhardt, Q. Liu, A. Lamas-Linares, J. Skaar, C. Kurtsiefer, and V. Makarov, “Full-field implementation of a perfect eavesdropper on a quantum cryptography system,” *Nature Communications*, vol. 2, p. 349, 2011.
- [15] J. Dynes *et al.*, “Cambridge quantum network,” *npj Quantum Information*, vol. 5, 12 2019.
- [16] W. Yan, X. Zheng, W. Wen, L. Lu, Y. Du, Y. Lu, S. Zhu, and X.-S. Ma, “A measurement-device-independent quantum key distribution network using optical frequency comb,” 2025.
- [17] H. Qin *et al.*, “The National Quantum-Safe Network in Singapore,” in *Proceedings of QCrypt 2024*, 2024.
- [18] M. Pittaluga *et al.*, “Coherent Quantum Communications Across National Scale Telecommunication Infrastructure,” 2024.
- [19] Y.-L. Tang *et al.*, “Measurement-Device-Independent Quantum Key Distribution over Untrustful Metropolitan Network,” *Physical Review X*, vol. 6, Mar. 2016.
- [20] R. C. Berrevoets *et al.*, “Deployed measurement-device independent quantum key distribution and Bell-state measurements coexisting with standard internet data and networking equipment,” *Communications Physics*, vol. 5, July 2022.
- [21] Q*Bird, “Fortifying Europe’s quantum communication – Q*Bird and BeQCI achieve the first cross-border MDI QKD link in Benelux,” 2025.
- [22] V. Martin *et al.*, “MadQCI: a heterogeneous and scalable SDN-QKD network deployed in production facilities,” *npj Quantum Information*, vol. 10, 09 2024.
- [23] E. Commission, “Open European Quantum Key Distribution Testbed,” 2024.
- [24] R. Yehia, S. Neves, E. Diamanti, and I. Kerenidis, “Quantum City: simulation of a practical near-term metropolitan quantum network,” 2022.
- [25] V. Karavias, E. Hugues-Salas, A. Beghelli, A. Lord, and M. C. Payne, “Comparative Cost Analysis of Prepare and Measure and Entanglement-Based QKD Networks,” in *International Conference on Optical Network Design and Modeling, ONDM, 2025*, 2025.

- [26] T. Coopmans *et al.*, “NetSquid, a NETwork Simulator for QUantum Information using Discrete events,” *Communications Physics*, vol. 4, no. 1, 2021.
- [27] T. R. Henderson, M. Lacage, G. F. Riley, C. Dowell, and J. Kopena, “Network simulations with the ns-3 simulator,” in *SIGCOMM demonstration*, vol. 14, p. 527, 2008.
- [28] A. Varga, “The OMNeT++ discrete event simulation system,” in *Proceedings of the European Simulation Multiconference (ESM’2001)*, vol. 9, p. 65, 2001.
- [29] P. W. Shor and J. Preskill, “Simple Proof of Security of the BB84 Quantum Key Distribution Protocol,” *Physical Review Letters*, vol. 85, p. 441–444, July 2000.
- [30] G. Brassard, N. Lütkenhaus, T. Mor, and B. C. Sanders, “Limitations on Practical Quantum Cryptography,” *Phys. Rev. Lett.*, vol. 85, pp. 1330–1333, Aug 2000.
- [31] H.-K. Lo, X. Ma, and K. Chen, “Decoy State Quantum Key Distribution,” *Phys. Rev. Lett.*, vol. 94, p. 230504, Jun 2005.
- [32] C. K. Hong, Z. Y. Ou, and L. Mandel, “Measurement of subpicosecond time intervals between two photons by interference,” *Phys. Rev. Lett.*, vol. 59, pp. 2044–2046, Nov. 1987.
- [33] S. Pirandola, R. Laurenza, C. Ottaviani, and L. Banchi, “Fundamental limits of repeaterless quantum communications,” *Nature Communications*, vol. 8, Apr. 2017.
- [34] M. Lucamarini, Z. L. Yuan, J. F. Dynes, and A. J. Shields, “Overcoming the rate–distance limit of quantum key distribution without quantum repeaters,” *Nature*, vol. 557, p. 400–403, May 2018.
- [35] Corning, “SMF-28 Ultra Optical Fiber – Product Information Sheet PI-1424-AEN.” Product datasheet, 2023.
- [36] ID Quantique, “ID281 Superconducting Nanowire Single-Photon Detection System – Product Datasheet.” Product datasheet, 2024.
- [37] OZ Optics, “Polarization Maintaining Fused Fiber Couplers/Splitters – Product Datasheet (DTS0092).” Product datasheet, 2013.
- [38] A. Duplinskiy, V. Ustimchik, A. Kanapin, V. Kurochkin, and Y. Kurochkin, “Low loss QKD optical scheme for fast polarization encoding,” *Optics Express*, vol. 25, no. 23, pp. 28886–28897, 2017.

- [39] Quandela, “Prometheus Single-Photon Source – Product Datasheet.” Product datasheet, 2024.
- [40] F. Grasselli *et al.*, “Quantum Key Distribution with Basis-Dependent Detection Probability,” *Physical Review Applied*, vol. 23, p. 044011, 2025.
- [41] Thorlabs, “TN1550R5F2 – Narrowband Fiber Optic Coupler, 1550 nm, 50:50 Ratio – Product Datasheet (TTN168601-S01, Rev C).” Product datasheet, 2021.
- [42] G. P. Agrawal, *Fiber-Optic Communication Systems*. Wiley Series in Microwave and Optical Engineering, Wiley, 4th ed., 2010.
- [43] ID Quantique, “ID230 Infrared Single-Photon Detector – Product Datasheet.” Product datasheet, 2024.
- [44] E. Waks, C. Santori, and Y. Yamamoto, “Security aspects of quantum key distribution with a photon number resolving detector,” *Physical Review A*, vol. 66, no. 4, p. 042315, 2002.
- [45] Q*Bird, “Falqon® Series,” 2026. Accessed 2026-08-22.
- [46] C. Elliott, “Building the quantum network,” *New Journal of Physics*, vol. 4, p. 46, July 2002.
- [47] A. Lord, R. Woodward, S. Murai, H. Sato, J. Dynes, P. Wright, C. White, R. Davey, M. Wilkinson, P. Clinton-Tarestad, I. Hawkins, K. Farrington, and A. Shields, “London Quantum-Secured Metro Network,” 2023.
- [48] H. W. Kuhn, “A note on Fermat’s problem,” *Mathematical Programming*, vol. 4, no. 1, pp. 98–107, 1973.
- [49] S. R. Finch, *Mathematical Constants*, vol. 94 of *Encyclopedia of Mathematics and its Applications*. Cambridge University Press, 2003. Section 8.1, Geometric Probability Constants, pp. 479–484.
- [50] A. Acín, N. Brunner, N. Gisin, S. Massar, S. Pironio, and V. Scarani, “Device-Independent Security of Quantum Cryptography against Collective Attacks,” *Phys. Rev. Lett.*, vol. 98, p. 230501, Jun 2007.
- [51] B.-W. Lu *et al.*, “Device-independent quantum key distribution over 100 km with single atoms,” *Science*, vol. 391, no. 6785, pp. 592–597, 2026.

- [52] T.-Y. Chen, X. Jiang, S.-B. Tang, *et al.*, “Implementation of a 46-node quantum metropolitan area network,” *npj Quantum Information*, vol. 7, p. 134, Sept. 2021.
- [53] R. Yehia, Y. Piétri, C. Pascual-García, P. Lefebvre, and F. Centrone, “Energetic Analysis of Emerging Quantum Communication Protocols,” 2025.
- [54] N. Lütkenhaus, “Security against individual attacks for realistic quantum key distribution,” *Phys. Rev. A*, vol. 61, p. 052304, Apr 2000.
- [55] D. Gottesman, H.-K. Lo, N. Lütkenhaus, and J. Preskill, “Security of quantum key distribution with imperfect devices,” *Quantum Information & Computation*, vol. 4, no. 5, pp. 325–360, 2004.
- [56] E. Y.-Z. Tan, R. Schwonnek, K. T. Goh, I. W. Primaatmaja, and C. C.-W. Lim, “Computing secure key rates for quantum cryptography with untrusted devices,” *npj Quantum Information*, vol. 7, p. 158, 2021.
- [57] T.-T. Song, Q.-Y. Wen, F.-Z. Guo, and X.-Q. Tan, “Finite-key analysis for measurement-device-independent quantum key distribution,” *Phys. Rev. A*, vol. 86, p. 022332, Aug 2012.
- [58] H.-J. Briegel, W. Dür, J. I. Cirac, and P. Zoller, “Quantum Repeaters: The Role of Imperfect Local Operations in Quantum Communication,” *Physical Review Letters*, vol. 81, pp. 5932–5935, Dec. 1998.
- [59] K. Azuma, K. Tamaki, and H.-K. Lo, “All-photonic quantum repeaters,” *Nature Communications*, vol. 6, Apr. 2015.
- [60] H. J. Kimble, “The quantum internet,” *Nature*, vol. 453, p. 1023–1030, June 2008.
- [61] W. Miehle, “Link-Length Minimization in Networks,” *Operations Research*, vol. 6, no. 2, pp. 232–243, 1958.
- [62] M. E. O’Kelly, “The Location of Interacting Hub Facilities,” *Transportation Science*, vol. 20, no. 2, pp. 92–106, 1986.
- [63] S. Alumur and B. Y. Kara, “Network hub location problems: The state of the art,” *European Journal of Operational Research*, vol. 190, no. 1, pp. 1–21, 2008.